Rupadarshi's MS thesis notebook

Short link: https://dub.sh/thesisms
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Summary holidarys will start from Wed 7 May, '25 and end on Mon 4 Aug, '25.

4 days left in summer holidays of 90 days

95.60% complete

MS thesis period will start from Mon 4 Aug, '25 and end on Sat 4 Apr, '26 tentatively.

4 days to go

FYI: It is a 244 days of MS thesis year.

notes

- *ergodic dynamics*
  - Measure preserving endomorphisms
  - Recurrence in iterations of a measure preserving endomorphism
  - Interpretation of measure preserving endomorphisms
  - ergodic theorems
  - spectral theory and ergodicity
  - Haar measure on locally compact groups
  - Actions of locally compact groups
- *hyperbolic surfaces*
  - $SL(2,\mathbb{R}) \curvearrowright \mathbb{R}H^2$
  - geodesic flow on hyperbolic surfaces is ergodic

notes on references

- articles, papers
  - Coifman and Weiss - Transference Methods in Analysis
  - Amos Nevo - Pointwise Ergodic Theorems for Actions of Groups
- courses, lecture notes
  - Uri Bader - A course in linear groups and ergodic theory
  - sub-Riem_notes - 2024 - September.pdf
- books
  - Einsiedler and Ward - Ergodic Theory
  - Sigurdur Helgason - Groups and geometric analysis

- Eisner, Farkas, Haase, Nagel - Operator Theoretic Aspects of Ergodic Theory-Springer (2015).pdf
- Amie Wilkinson - Geometry, Dynamics, and Rigidity

order of presentations

- Presentation 1 (19 June '25)
  - singular integral transforms on $\mathbb{R}^n$, transference methods 1
- Meet (26 June '25)
  - Ergodic transformations, *spectral invariants* of ergodic transformations
- Presentation 2 (10 July '25)
  - Amenable groups
- Meet (24 July '25)
  - *quantum (unique) ergodicity*
- Presentation 3 (31 July '25)
  - ergodic theorems
- Presentation 4
  - proof of ergodicity of geodesic flow
- Presentation 5
  - proof of ergodicity of hyperbolic linear maps on torus
- Presentation 6
- Presentation 7

1: Coifman and Weiss - Transference Methods in Analysis > singular integral transforms sections 1 and *stated* section 2

Measure preserving endomorphisms

Definition. Measure preserving endomorphisms

Let $(X, \mathcal{A}, \mu)$ be a $[0, \infty]$-measure space. Then a *measurable* map

$$T : X \to X$$

that **preserves the measure μ** that is

$$A \in \mathcal{A} \implies \mu(T^{-1}(A)) = \mu(A)$$

Here, μ is called a T-invariant measure and T is called μ-preserving transformation.

Definition. Ergodic endomorphisms

A μ-preserving transformation $T : X \to X$ on a measure space $(X, \mathcal{A}, \mu)$ is called **μ-ergodic** if

$$A \in \mathcal{A}, T^{-1}(A) = A \implies \mu(A) = 0 \text{ or } \mu(X \setminus A) = 0$$

Here, μ is called an ergodic measure for T.

Definition. Strongly mixing endomorphisms

A μ-preserving transformation $T : X \to X$ on a **probability** space $(X, \mathcal{A}, \mu)$ is called **strongly mixing** if

$$A, B \in \mathcal{A} \implies \mu(A \cap T^{-n}(B)) \xrightarrow{n\to\infty} \mu(A)\mu(B)$$

Strong mixing on a probability space $\implies$ ergodicity.

Suppose $A \in \mathcal{A}$ such that $T^{-1}(A) = A$. Then

$$\mu(E) = \mu(E \cap T^{-n}(E)) \xrightarrow{n\to\infty} \mu(E)^2$$

which means

$$\mu(E)^2 = \mu(E) \implies \mu(E) = 1 \text{ or } 0$$

Recurrence

Recurrence in iterations of a measure preserving endomorphism

[1]

- Let $(X, \mathcal{A}, \mu)$ be a **probability space** and $T : X \to X$ be a μ-preserving transformation. For a measurable subset $A \subseteq X$, consider the subsets

$$A, T^{-1}(A), T^{-2}(A), \ldots$$

  which are all measurable subsets of X.
  - As

$$T^{-n}(A) \subseteq X$$

    we have

$$\bigcup_{n \in \mathbb{N}} T^{-n}(A) \subseteq X$$
$$\implies \mu\Big(\bigcup_{n \in \mathbb{N}} T^{-n}(A)\Big) \leq \mu(X) < \infty$$

  - But as $\mu(A) = \mu(T^{-n}(A))$, if $T^{-n}(A)$ are all disjoint sets we have

$$\mu(X) \geq \sum_{n \geq 0} \mu(T^{-1}(A)) = \sum_{n \geq 0} \mu(A)$$

  - Hence we conclude

$$\mu(A) > 0 \implies \{T^{-n}(A) \mid n \in \mathbb{N}\} \text{ are not disjoint}$$

    which is $\iff$ there exists $n, m \in \mathbb{N}, n > m$

$$x \in T^{-n}(A) \cap T^{-m}(A) \implies T^n(x) \in AT^{m-n}(T^n x) = T^m(x) \in A$$

    that is there exists points in A return back to A.
  - The *contrapositive* of the above statement is

$$\{T^{-n}(A) \mid n \in \mathbb{N}\} \text{ are disjoint} \implies \mu(A) = 0$$

- Thus for a subset $B \subseteq X$ with $\mu(B) > 0$, there are points in $B \cap T^{-n}(B)$ for some n, that means there are points in B that return back to B after some iterations.
- Consider the set $A \subset B$ of points that do not return to B

$$A := \{x \in B \mid \forall n > 0 : T^n(x) \notin B\}$$

  , so this means points of A do not return to A either. If $T^{-n}(A)$ are *not* pairwise disjoint for all $n \in \mathbb{N}$ then there exists $n, m \in \mathbb{N}, n > m$

$$x \in T^{-n}(A) \cap T^{-m}(A) \implies T^n(x) \in A, T^{m-n}(T^n x) = T^m(x) \in A$$

  meaning the point $T^n(x)$ in A returns to A, which contradicts the definition of A. Thus $T^{-n}(A)$ are all disjoint.

Thus applying the same conclusion for the set of points in B that do *not* return back to B we show that it is a set of measure zero:

(weak form of Poincare recurrence) **Let $(X, \mathcal{A}, \mu)$ be a probability space and $T : X \to X$ be a μ-preserving transformation and $B \subseteq X$ such that $\mu(B) > 0$. Then the set of points in B that never returns to B has measure zero.**

- - Consider the set $A \subset B$ of points that do not return to B

    $$A := \{x \in B \mid \forall n > 0 : T^n(x) \notin B\}$$

    , so this means points of A do not return to A either. If $T^{-n}(A)$ are *not* pairwise disjoint for all $n \in \mathbb{N}$ then there exists $n, m \in \mathbb{N}, n > m$

    $$x \in T^{-n}(A) \cap T^{-m}(A) \implies T^n(x) \in A, T^{m-n}(T^n x) = T^m(x) \in A$$

    meaning the point $T^n(x)$ in A returns to A, which contradicts the definition of A. Thus $T^{-n}(A)$ are all disjoint.
- By
  - Let $(X, \mathcal{A}, \mu)$ be a **probability space** and $T : X \to X$ be a μ-preserving transformation. For a measurable subset $A \subseteq X$, consider the subsets

    $$A, T^{-1}(A), T^{-2}(A), \dots$$

    which are all measurable subsets of X.
    - As

      $$T^{-n}(A) \subseteq X$$

      we have

      $$\begin{aligned} \bigcup_{n \in \mathbb{N}} T^{-n}(A) &\subseteq X \\ \implies \mu\left(\bigcup_{n \in \mathbb{N}} T^{-n}(A)\right) &\leq \mu(X) < \infty \end{aligned}$$

    - But as $\mu(A) = \mu(T^{-n}(A))$, if $T^{-n}(A)$ are all disjoint sets we have

      $$\mu(X) \geq \sum_{n \geq 0} \mu(T^{-1}(A)) = \sum_{n \geq 0} \mu(A)$$

    - Hence we conclude

      $$\mu(A) > 0 \implies \{T^{-n}(A) \mid n \in \mathbb{N}\} \text{ are not disjoint}$$

      which is $\iff$ there exists $n, m \in \mathbb{N}, n > m$

      $$x \in T^{-n}(A) \cap T^{-m}(A) \implies T^n(x) \in AT^{m-n}(T^n x) = T^m(x) \in A$$

      that is there exists points in A return back to A.
    - The *contrapositive* of the above statement is

      $$\{T^{-n}(A) \mid n \in \mathbb{N}\} \text{ are disjoint} \implies \mu(A) = 0$$

  we conclude

  $$\{T^{-n}(A) \mid n \in \mathbb{N}\} \text{ are all disjoint} \implies \mu(A) = 0$$

- Now consider the set of points in B that do not return to B *infinitely many times* that is

$$A_\infty := \{x \in B \mid \exists k \geq 1 : \forall n > k(T^n(x) \notin B)\}$$

  - If we consider the sets

$$A := \{x \in B \mid \forall n > 0(T^n(x) \notin B)\}$$
$$A_k := \{x \in B \mid T^k(x) \in B \text{ and } \forall n > k(T^n(x) \notin B)\}$$

    for $k \geq 1$,
  - We see

$$x \in T^{-k}(A_0) \iff T^k(x) \in A$$
$$\iff T^k(x) \in B \text{ and } \forall n > 0 \;\; T^n(T^k x) \notin B$$

    so

$$A_k = T^{-k}(A)$$

    for all $k \geq 1$.
  - If $x \in A_\infty$ there must be a largest integer k such that $T^k(x) \in B$ so $x \in A_k$. Conversely, if $x \in B \cap A_k$, we know

$$T^n(x) \notin B$$

    for all $n > k$ meaning $x \in A_\infty$.
  - Thus

$$A_\infty = B \cap \bigcup_{k \in \mathbb{N} \cup \{\infty\}} A_k$$

    implying

$$A_\infty \subseteq \bigcup_{k \in \mathbb{N} \cup \{\infty\}} A_k$$
$$\implies \mu(A_\infty) \leq \mu\left(\bigcup_{k \in \mathbb{N} \cup \{\infty\}} A_k\right)$$

  - And we have already proven $\{A_k \mid k \in \mathbb{N}\}$ are all disjoint
    - Consider the set $A \subset B$ of points that do not return to B

$$A := \{x \in B \mid \forall n > 0 : T^n(x) \notin B\}$$

      , so this means points of A do not return to A either. If $T^{-n}(A)$ are *not* pairwise disjoint for all $n \in \mathbb{N}$ then there exists $n, m \in \mathbb{N}, n > m$

$$x \in T^{-n}(A) \cap T^{-m}(A) \implies T^n(x) \in A, T^{m-n}(T^n x) = T^m(x) \in A$$

      meaning the point $T^n(x)$ in A returns to A, which contradicts the definition of A. Thus $T^{-n}(A)$ are all disjoint.
- Thus we can conclude by the previous argument

$$\mu(A_k) = 0 \implies \mu(A_\infty) \leq \sum_{k \in \mathbb{N} \cup \{\infty\}} \mu(A_k) = 0$$

(infinite Poincare recurrence) Let $(X, \mathcal{A}, \mu)$ be a probability space and $T : X \to X$ be a μ-preserving transformation and $B \subseteq X$ such that $\mu(B) > 0$. Then the set of points in B that returns to B *only finitely many times* has measure zero.

- Now consider the set of points in B that do not return to B *infinitely many times* that is

$$A_\infty := \{x \in B \mid \exists k \geq 1 : \forall n > k(T^n(x) \notin B)\}$$

  - If we consider the sets

$$\begin{aligned} A &:= \{x \in B \mid \forall n > 0(T^n(x) \notin B)\} \\ A_k &:= \{x \in B \mid T^k(x) \in B \text{ and } \forall n > k(T^n(x) \notin B)\} \end{aligned}$$

    for $k \geq 1$,
  - We see

$$\begin{aligned} x \in T^{-k}(A_0) &\iff T^k(x) \in A \\ &\iff T^k(x) \in B \text{ and } \forall n > 0 \ \ T^n(T^k x) \notin B \end{aligned}$$

    so

$$A_k = T^{-k}(A)$$

    for all $k \geq 1$.
  - If $x \in A_\infty$ there must be a largest integer k such that $T^k(x) \in B$ so $x \in A_k$. Conversely, if $x \in B \cap A_k$, we know

$$T^n(x) \notin B$$

    for all $n > k$ meaning $x \in A_\infty$.
  - Thus

$$A_\infty = B \cap \bigcup_{k \in \mathbb{N} \cup \{\infty\}} A_k$$

    implying

$$\begin{aligned} A_\infty &\subseteq \bigcup_{k \in \mathbb{N} \cup \{\infty\}} A_k \\ \implies \mu(A_\infty) &\leq \mu\Bigg(\bigcup_{k \in \mathbb{N} \cup \{\infty\}} A_k\Bigg) \end{aligned}$$

  - And we have already proven $\{A_k \mid k \in \mathbb{N}\}$ are all disjoint
    - Consider the set $A \subset B$ of points that do not return to B

$$A := \{x \in B \mid \forall n > 0 : T^n(x) \notin B\}$$

      , so this means points of A do not return to A either. If $T^{-n}(A)$ are *not* pairwise disjoint for all $n \in \mathbb{N}$ then there exists $n, m \in \mathbb{N}, n > m$

$$x \in T^{-n}(A) \cap T^{-m}(A) \implies T^n(x) \in A, T^{m-n}(T^n x) = T^m(x) \in A$$

      meaning the point $T^n(x)$ in A returns to A, which contradicts the definition of A. Thus $T^{-n}(A)$ are all disjoint.

- Let $(X, \mathcal{A}, \mu)$ be a **probability space** and $T : X \to X$ be a μ-preserving transformation. For a measurable subset $A \subseteq X$, consider the subsets

$$A, T^{-1}(A), T^{-2}(A), \ldots$$

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- As

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we have

$$\begin{aligned} \bigcup_{n\in\mathbb{N}} T^{-n}(A) \subseteq X \\ \implies \mu\left(\bigcup_{n\in\mathbb{N}} T^{-n}(A)\right) \leq \mu(X) < \infty \end{aligned}$$

- But as $\mu(A) = \mu(T^{-n}(A))$, if $T^{-n}(A)$ are all disjoint sets we have

$$\mu(X) \geq \sum_{n\geq 0} \mu(T^{-1}(A)) = \sum_{n\geq 0} \mu(A)$$

- Hence we conclude

$$\mu(A) > 0 \implies \{T^{-n}(A) \mid n \in \mathbb{N}\} \text{ are not disjoint}$$

which is $\iff$ there exists $n, m \in \mathbb{N}, n > m$

$$x \in T^{-n}(A) \cap T^{-m}(A) \implies T^n(x) \in AT^{m-n}(T^n x) = T^m(x) \in A$$

that is there exists points in A return back to A.

- The *contrapositive* of the above statement is

$$\{T^{-n}(A) \mid n \in \mathbb{N}\} \text{ are disjoint} \implies \mu(A) = 0$$

- Thus we can conclude by the previous argument

$$\mu(A_k) = 0 \implies \mu(A_\infty) \leq \sum_{k\in\mathbb{N}\cup\{\infty\}} \mu(A_k) = 0$$

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1. C3L3N.pdf ↩

Interpretation of measure preserving endomorphisms

X	$x \in X$	
$B \subseteq X$	$x \in B$	
χ_B	$\chi_B(x)$	$\mathtt{check}(x \in B) \in \{0,1\}$
μ	$\mu(B)$	$\mu(B) = \mu\text{–}\mathtt{P}(x \in B)$ is the **size** of B with respect to μ is the **probability** of randomly picking a point x in B
$T : X \to X$ that is μ-invariant	**Definition. Measure preserving endomorphisms** Let $(X, \mathcal{A}, \mu)$ be a $[0,\infty]$-measure space. Then a *measurable* map $T : X \to X$ that **preserves the measure** μ that is $A \in \mathcal{A} \implies \mu(T^{-1}(A)) = \mu(A)$ Here, μ is called a T-invariant measure and T is called μ-preserving transformation.	$\mu(T^{-1}(A)) = \mu(A)$ the **size** of *all* points going into A after one iteration is same as the **size** of A
		$\mu\text{–}\mathtt{P}(x \in B) = \mu\text{–}\mathtt{P}(Tx \in B)$ **probability** of picking x which is in B after an iteration $=$ **probability** of picked x in B
	$f : X \to \mathbb{R}$	*observable*
	$U_T(f) := f \circ T : X \to \mathbb{R}$	
	$\mathfrak{X}_T^{(n)} := f \circ T^n : X \to \mathbb{R}$ a stochastic process?	
	$\int_X f$	$\mu\text{–}\mathtt{E}(f)$ *expectation value*
	$\mu(B) = \int_X \chi_B \mathrm{d}\mu$	$\mu(B) = \mu\text{–}\mathtt{E}(\mathtt{check}(- \in B))$

$\chi_B \circ T^k$	$\chi_B \circ T^k(x)$	$\texttt{check}(T^k(x) \in B) \in \{0,1\}$
		$\texttt{returns}(B) := \{x \in B \vert \exists n > 0 : T^n(x)$
	• Hence we conclude $\mu(A) > 0 \implies \{T^{-n}(A) \mid n \in$ which is $\iff$ there exists $n, m \in \mathbb{N}, n > m$ $x \in T^{-n}(A) \cap T^{-m}(A) \implies T$ that is there exists points in A return back to A.	$\mu(B) > 0 \implies \texttt{returns}(B) \neq \emptyset$ or $\texttt{returns}(B) = \emptyset \implies \mu(B) = 0$
	• Consider the set $A \subset B$ of points that do not return to B $A := \{x \in B \mid \forall n > 0 : T^n(x) \notin$, so this means points of A do not return to A either. If $T^{-n}(A)$ are *not* pairwise disjoint for all $n \in \mathbb{N}$ then there exists $n, m \in \mathbb{N}, n > m$ $x \in T^{-n}(A) \cap T^{-m}(A) \implies T$ meaning the point $T^n(x)$ in A returns to A, which contradicts the definition of A. Thus $T^{-n}(A)$ are all disjoint.	$\texttt{returns}(B \setminus \texttt{returns}(B)) = \emptyset$
	(weak form of Poincare recurrence) Let $(X, \mathcal{A}, \mu)$ be a probability space and $T : X \to X$ be a μ-preserving transformation and $B \subseteq X$ such that $\mu(B) > 0$. Then the set of points in B that never returns to B has measure zero.	$\mu(B) > 0$ $\implies \mu(B \setminus \texttt{returns}(B)) = 0$

	(infinite Poincare recurrence) Let $(X, \mathcal{A}, \mu)$ be a probability space and $T : X \to X$ be a μ-preserving transformation and $B \subseteq X$ such that $\mu(B) > 0$. Then the set of points in B that returns to B *only finitely many times* has measure zero.	
ergodic	Definition. **Ergodic endomorphisms** A μ-preserving transformation $T : X \to X$ on a measure space $(X, \mathcal{A}, \mu)$ is called **μ-ergodic** if $$A \in \mathcal{A}, T^{-1}(A) = A \implies \mu(A) =$$ Here, μ is called an ergodic measure for T.	
strongly mixing	Definition. **Strongly mixing endomorphisms** A μ-preserving transformation $T : X \to X$ on a **probability** space $(X, \mathcal{A}, \mu)$ is called **strongly mixing** if $$A, B \in \mathcal{A} \implies \mu(A \cap T^{-n}(B)) \xrightarrow{n}$$	μ–$\mathtt{P}(x \in A \& T^n x \in B) \to \mu(A)\mu(B)$

ergodic theorems

mean ergodic theorem

Example

$$T(x) = x + 1$$
$$U_T(f)(x) = f(x+1)$$
$$A_T^{(n)}(f) = \frac{1}{n} \sum f(x+k)$$

Proposition: Let $(X, \mathfrak{M}, \mu)$ be a measure space, $T : X \to X$ be a μ-preserving transformation and

$$P_T : L^2(X, \mu) \to L^2(X, \mu)$$

be the orthogonal projection onto the close subspace

$$\{f \in L^2(X, \mu) \big| U_T f = f\} = \ker(U_T - \mathsf{Id})$$

- Then

$$\mathrm{im}\,(U_T - \mathsf{Id})^{\perp} = \ker(U_T - \mathsf{Id})$$

- This implies

$$L^2(X, \mu) = \ker(U - \mathsf{Id}) \oplus \overline{\mathrm{im}\,(U_T - \mathsf{Id})}$$

  so $f = P_T f + h$ for $h \in \overline{\mathrm{im}\,(U_T - \mathsf{Id})}$ where

$$A_T^{(n)}(P_T f) = P_T f$$

- $$A_T^{(n)}(h) \xrightarrow{n \to \infty, L^2} 0$$

  Thus we conclude

$$A_T^{(n)}(f) \xrightarrow{n \to \infty, L^2} P_T(f)$$

- For $f = U_T g - g \in \mathrm{im}\,(U_T - \mathsf{Id})$ we have

$$\begin{aligned} \frac{1}{n} \sum_{k=0}^{n-1} U_T^k f &= \frac{1}{n} \sum_{k=0}^{n-1} U_T^k (U_T g - g) \\ &= \frac{1}{n}(U_T g - g + U_T^2 - U_T g + \cdots + U_T^n g - U_T^{n-1} g) \\ &= \frac{1}{n}(U_T^n g - g) \\ \left\| \frac{1}{n} \sum_{k=0}^{n-1} U_T^k f \right\|_2 &= \frac{1}{n} \|U_T^n g - g\|_2 \xrightarrow{n \to \infty} 0 \end{aligned}$$

  Thus

$$f \in B \implies A_T^{(n)}(f) \xrightarrow{n \to \int, L^2} 0$$

-

$T : X \to X$	
$U_T : L^2(X,\mu) \to L^2(X\mu)$	
$A_T^{(n)} : L^2(X,\mu) \to L^2(X,\mu)$	
P_T	

(Mean ergodic theorem for measure preserving transformations) Let $(X, \mathfrak{M}, \mu)$ be a measure space, $T : X \to X$ be a μ-preserving transformation. Then

$$A_T^{(n)} := \frac{1}{n}\sum_{k=0}^{n-1} U_T^k \xrightarrow{n\to\infty,\text{weak* } L^2} P_T$$

Proposition: Let $(X, \mathfrak{M}, \mu)$ be a measure space, $T : X \to X$ be a μ-preserving transformation and

$$P_T : L^2(X,\mu) \to L^2(X,\mu)$$

be the orthogonal projection onto the close subspace

$$\{f \in L^2(X,\mu) \big| U_T f = f\} = \ker(U_T - \mathsf{Id})$$

- Then

$$\text{im } (U_T - \mathsf{Id})^\perp = \ker(U_T - \mathsf{Id})$$

- This implies

$$L^2(X,\mu) = \ker(U - \mathsf{Id}) \oplus \overline{\text{im } (U_T - \mathsf{Id})}$$

so $f = P_T f + h$ for $h \in \overline{\text{im } (U_T - \mathsf{Id})}$ where

$$A_T^{(n)}(P_T f) = P_T f$$

- $$A_T^{(n)}(h) \xrightarrow{n\to\infty, L^2} 0$$

Thus we conclude

$$A_T^{(n)}(f) \xrightarrow{n\to\infty, L^2} P_T(f)$$

Proposition: Let (X, μ) be a probability space.

$$f \in L^1(X,\mu) \implies A_T^{(n)}(f) \xrightarrow{n\to\infty, L^2} \bar{f} \in L^1(X,\mu)^T$$

- As (X,μ) is a probability space, $L^\infty \subset L^2$. Now for any $g \in L^\infty \subset L^2$ by

> **(Mean ergodic theorem for measure preserving transformations)** Let $(X, \mathfrak{M}, \mu)$ be a measure space, $T : X \to X$ be a μ-preserving transformation. Then
>
> $$A_T^{(n)} := \frac{1}{n}\sum_{k=0}^{n-1} U_T^k \xrightarrow{n\to\infty,\text{weak}^*\ L^2} P_T$$

we have

$$A_T^{(n)}(g) \xrightarrow{n\to\infty, L^2} g'$$

- Because

$$\begin{aligned} &\left\|A^{(n)}(g)\right\|_\infty \le \|g\|_\infty \\ \implies &\left|\left\langle A_T^{(n)}(g), \chi_B\right\rangle\right| \le \|g\|_\infty \mu(B) \\ &\qquad\Big\downarrow{\scriptstyle n\to\infty} \\ &|\langle g', \chi_B\rangle| \le \|g\|_\infty \mu(B) \end{aligned}$$

maximal ergodic theorem

Let $(X, \mathfrak{M}_X, \mu)$ be a probability space, $T : X \to X$ be μ-preserving and $B \in \mathfrak{M}_X$.

event/function	value	
χ_B	$\chi_B(x)$	$\texttt{check}(x \in B) \in \{0,1\}$
	$\mu(B) = \int_X \chi_B \mathrm{d}\mu$	$\mu(B) = \mu\text{–}\mathtt{P}(B) = \mu\text{–}\mathtt{E}(\texttt{check}(- \in B)$
$\chi_B \circ T^k$	$\chi_B \circ T^k(x)$	$\texttt{check}(T^k(x) \in B) \in \{0,1\}$
$A_T^{(n)}(\chi_B)$	$A_T^{(n)}(\chi_B)(x)$	$\texttt{average}\left\{\texttt{check}(T^k(x) \in B) \middle\| 0 \le k <$ $=: \texttt{time\_spent}(x, B, n) \in [0,1]$ is the *average time spent* by x in B until n
$A_T^{(n)}(\chi_B) > \alpha$		$\{x \| \texttt{time\_spent}(x, B, n) > \alpha\}$ is the set of all points in X that spends more than α average time in B until n
	$\mu(A_T^{(n)}(\chi_B) > \alpha)$ $\le \frac{1}{\alpha}\int_X A_T^{(n)}(\chi(B))\mathrm{d}\mu$ $\le \frac{1}{\alpha}\mu(B)$	$\mu\{x \| \texttt{time\_spent}(x, B, n) > \alpha\}$ $\le \frac{1}{\alpha}\mu\text{–}\mathtt{E}(A_T^{(n)}(\chi_B))$ $\le \frac{1}{\alpha}\mu(B)$

event/function	value	
If $\mu(B_\epsilon) = \epsilon \in (0,1)$ and $\alpha = \sqrt{\epsilon} > \epsilon$	$\mu(A_T^{(n)}(\chi_{B_\epsilon}) < \sqrt{\epsilon}) \leq \sqrt{\epsilon}$	If $\mu(B_\epsilon) = \epsilon$ we have $\mu(\texttt{who\_spent}(B_\epsilon, \sqrt{\epsilon}, n)) \leq \sqrt{\epsilon}$ that is, the **size** of set of *all* points that spend $\sqrt{\epsilon}$ proportion of its future until n inside B is $\leq \sqrt{\epsilon}$
(Maximal ergodic theorem for measure preserving transformations) Let $(X, \mathfrak{M}_X, \mu)$ be a probability space, $T : X \to X$ be μ-preserving and $f \in L^1(X, \mu)$ be $\mathbb{R}$-valued. Then for $\alpha \in \mathbb{R}$ the set $E_\alpha = \left\{ x \in X \mid \sup_{n \geq 1} \frac{1}{n} \sum_{k=0} f(T^i \right.$ **has measure** $\alpha\mu(E_\alpha) \leq \int_{E_\alpha} f \mathrm{d}\mu \leq \lVert f \rVert_1$ **Moreover** $T^{-1}A = A \implies \alpha\mu(E_\alpha \cap A)$	for $g = \chi_{B_\epsilon}, E_{\sqrt{\epsilon}} =: B'_\epsilon$ we have $\mu(B'_\epsilon) \leq \epsilon$	for $\mu(B_\epsilon) = \epsilon$ we have $\texttt{who\_spent}(B_\epsilon, \sqrt{\epsilon}) = B$ $:= \left\{ x \mid \sup_{n\geq 1} \texttt{time\_spent}(x, B, n) > \sqrt{\epsilon} \right\}$ then and $x \in X \setminus B'_\epsilon$ $\forall n : \texttt{time\_spent}(x, B, n) \leq \epsilon$ thus the **size** of all points that spend $\sqrt{\epsilon}$ proportion of its future inside B_ϵ (of size ϵ) is $\leq \sqrt{\epsilon}$ $\mu(\texttt{who\_spent}(B_\epsilon, \sqrt{\epsilon})) \leq \sqrt{\epsilon}$
(Pointwise ergodic theorem for measure preserving transformation) Let $(X, \mathfrak{M}_X, \mu)$ be a probability space, $T : X \to X$ be μ-preserving and $f \in L^1(X, \mu)$. Then $\langle f \rangle_T = \lim_{n\to\infty} A_T^{(n)}(f) = \lim_{n\to\infty} \frac{1}{n}$ **converges pointwise almost everywhere on X to a T-invariant function $\langle f \rangle_T \in L^1(X, \mu)^T$ and** $\int_X \langle f \rangle_T \mathrm{d}\mu = \int_X f \mathrm{d}\mu$		
	If T is ergodic then $A_T^{(n)}(\chi_B) \xrightarrow{n\to\infty,\mathrm{ae}} \mu(B)$	If T is ergodic then $\texttt{time\_spent}(x, B, n) \xrightarrow{n\to\infty} \mu(B)$ for almost every $x \in X$

(Maximal ergodic theorem for measure preserving transformations) **Let $(X, \mathfrak{M}_X, \mu)$ be a probability space, $T : X \to X$ be μ-preserving and $f \in L^1(X, \mu)$ be $\mathbb{R}$-valued. Then for $\alpha \in \mathbb{R}$, the set**

$$E_\alpha = \left\{ x \in X \middle| \sup_{n \geq 1} \frac{1}{n} \sum_{k=0} f(T^i x) > \alpha \right\}$$

has measure

$$\alpha \mu(E_\alpha) \leq \int_{E_\alpha} f \mathrm{d}\mu \leq \|f\|_1$$

Moreover

$$T^{-1}A = A \implies \alpha\mu(E_\alpha \cap A) \leq \int_{E_\alpha \cap A} f \mathrm{d}\mu$$

pointwise ergodic theorem

- Let $f : X \to \mathbb{R}$ be L^1.
- > **Proposition:** Let (X, μ) be a probability space.
  >
  > $$f \in L^1(X, \mu) \implies A_T^{(n)}(f) \xrightarrow{n\to\infty, L^2} \bar{f} \in L^1(X, \mu)^T$$

(Pointwise ergodic theorem for measure preserving transformation) **Let $(X, \mathfrak{M}_X, \mu)$ be a probability space, $T : X \to X$ be μ-preserving and $f \in L^1(X, \mu)$. Then**

$$\langle f \rangle_T = \lim_{n\to\infty} A_T^{(n)}(f) = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} f \circ T^k$$

converges pointwise almost everywhere on X to a T-invariant function $\langle f \rangle_T \in L^1(X, \mu)^T$ and

$$\int_X \langle f \rangle_T \mathrm{d}\mu = \int_X f \mathrm{d}\mu$$

counterexamples

Definition. **A continuous linear endomorphism $S : V \to V$ on a Hausdorff topological vector space V is mean ergodic if**

$$\frac{1}{n} \sum_{k=0}^{n-1} S^k$$

converges *pointwise* to the projection onto $\ker(S - \mathrm{Id})$

Definition. **A continuous lineat endomorphism $S : E \to E$ of a Banach space E is called power-bounded if**

$$\sup_{n\in\mathbb{N}} \|S^n\| < \infty$$

Every power-bounded linear endomorphism of a reflexive Banach space is mean ergodic.

[1]

(X,μ) is a $\to$	probability space	infinite positive measure space
	Proposition: Let (X,μ) be a probability space. If $p>q>0$ we have $L^p(X,\mu)\subset L^q(X,\mu)$ Thus, in particular $L^1\supset L^2\supset\cdots\supset L^\infty$	
mean ergodic in L^2	YES!	YES!
mean ergodic in L^1	YES? [2]	NO![3]
mean ergodic in L^p for $p\in(1,\infty)\setminus\{2\}$ **(reflexive)**	YES!	YES! Every power-bounded linear endomorphism of a reflexive Banach space is mean ergodic.
pointwise convergence ae of L^1	YES!	YES? [4]

---

1. Eisner, Farkas, Haase, Nagel - Operator Theoretic Aspects of Ergodic Theory-Springer (2015), p.149 ↩

Let $(\mathrm{X};\varphi)$ be a measure-preserving system. In von Neumann's theorem the Koopman operator was considered as an operator on $\mathrm{L}^2(\mathrm{X})$. However, it is natural to view it also as an operator on L^1 and in fact on any L^p-space with $1\le p\le\infty$.

Theorem 8.8. *Let $(\mathrm{X};\varphi)$ be a measure-preserving system, and consider the Koopman operator $T=T_\varphi$ on the space $\mathrm{L}^1=\mathrm{L}^1(\mathrm{X})$. Then*

$$P_T f := \lim_{n\to\infty}\frac{1}{n}\sum_{j=0}^{n-1}T^j f \qquad \text{exists in } \mathrm{L}^p \text{ for every } f\in\mathrm{L}^p,\ 1\le p<\infty.$$

Moreover, for $1\le p\le\infty$, the following assertions hold:

a) *The space $\mathrm{fix}(T)\cap\mathrm{L}^p$ is a Banach sublattice of L^p containing* $\mathbf{1}$.

b) *$P_T:\mathrm{L}^p\to\mathrm{fix}\,T\cap\mathrm{L}^p$ is a positive contractive projection satisfying*

$$\int_{\mathrm{X}} P_T f = \int_{\mathrm{X}} f \qquad (f\in\mathrm{L}^p) \tag{8.6}$$

$$\text{and}\qquad P_T(f\cdot P_T g)=(P_T f)\cdot(P_T g) \qquad (f\in\mathrm{L}^p,\ g\in\mathrm{L}^q,\ \tfrac{1}{p}+\tfrac{1}{q}=1). \tag{8.7}$$

2. ↩

3. https://mathoverflow.net/a/475916 ↩

Definition 11.3. Let X be a measure space and let $1 \leq p \leq \infty$. A bounded linear operator T on $\mathrm{L}^p(\mathrm{X})$ is called **pointwise ergodic** if the limit

$$\lim_{n\to\infty} \mathrm{A}_n f = \lim_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} T^j f$$

exists μ-almost everywhere for every $f \in \mathrm{L}^p(\mathrm{X})$.

Birkhoff's theorem says that Koopman operators arising from measure-preserving systems are pointwise ergodic. We shall obtain Birkhoff's theorem from a more general result which goes back to Hopf (1954) and Dunford and Schwartz (1958). Recall that an operator T on $\mathrm{L}^1(\mathrm{X})$ is called a **Dunford–Schwartz operator** (or an absolute contraction) if

$$\|Tf\|_1 \leq \|f\|_1 \quad \text{and} \quad \|Tf\|_\infty \leq \|f\|_\infty$$

for all $f \in \mathrm{L}^\infty \cap \mathrm{L}^1$. The Hopf–Dunford–Schwartz result then reads as follows.

Theorem 11.4 (Pointwise Ergodic Theorem). *Let* X *be a* general *measure space and let T be a positive Dunford–Schwartz operator on* $\mathrm{L}^1(\mathrm{X})$. *Then T is pointwise ergodic.*

4. ↩

spectral theory and ergodicity

spectral theorems

(Spectral theorem for bounded self-adjoint operators on separable Hilbert spaces, operator-valued integration version) If $A \in \mathcal{B}(H)$ is self-adjoint then $\exists!$ projection valued measure μ_A on the Borel space $\sigma(A) \subset \mathbb{C}$ with values in projections on H such that

$$\int_{\lambda\in\sigma(A)} \lambda \, \delta\mu_A = A$$

(Spectral theorem for bounded self-adjoint operators on separable Hilbert spaces, direct integration version) If $A \in \mathcal{B}(H)$ is self-adjoint then there $\exists$ σ-finite measure $\mu : \sigma(A) \to [0, \infty]$ and a unitary map

$$U : H \to \int^{\oplus}_{\lambda\in\sigma(A)} H_\lambda \, \delta\mu$$

such that $UAU^{-1}(s)(\lambda) = \lambda s(\lambda)$ for all sections s in the direct integral.

(Spectral theorem for bounded normal operators on separable Hilbert spaces, Hilbert bundle on the circle version) If $A \in \mathcal{B}(H)$ is normal then there exists a σ-finite? Borel measure $\mu : \sigma(A) \to [0, \infty]$, a Borel measurable

$$M : \sigma(A) \to \mathbb{N} \cup \{\infty\}$$

and a unitary isomorphism

$$U : H \cong L^2(E(M) \to S^1)$$

that sends A to the operator $A(f) = \lambda f(\lambda)$ where $E(M) \to S^1$ is the Hilbert bundle with fibres satisfying

$$\begin{cases} \dim E_\lambda = M(\lambda) & \\ E_\lambda \cong l^2(\mathbb{Z}) & \text{when } M(\lambda) = \infty \end{cases}$$

(Spectral theorem for bounded self-adjoint operators on separable Hilbert spaces, multiplier version) If $A \in \mathcal{B}(A)$ is self-adjoint then there exists a σ-finite measure space (X, μ), a bounded measurable $h : X \to \mathbb{R}$ and a unitary isomorphism

$$U : (H, A) \to (L^2(X, \mu), h)$$

Theorem 18.4 (Spectral Theorem, Multiplier Form). *For a bounded, normal operator T on a Hilbert space H, the pair (H, T) is unitarily equivalent to $(\mathrm{L}^2(\Omega, \mu), M)$, where μ is a positive Baire measure on a locally compact space Ω, and M is the multiplication operator associated with a $\sigma(T)$-valued continuous function on Ω.*

Proof. We employ the terminology introduced above. For each α let $K_\alpha := K \times \{\alpha\}$ be a copy of $K = \sigma(T)$, so that the sets K_α are pairwise disjoint. Let $\Omega := \bigcup_\alpha K_\alpha$ with the direct sum topology and the direct sum measure $\mu := \bigoplus_\alpha \mu_{x_\alpha}$. Then we obtain

$$H = \bigoplus_\alpha Z(x_\alpha) \cong \bigoplus_\alpha \mathrm{L}^2(K, \mu_{x_\alpha}) \cong \mathrm{L}^2\Big(\bigcup_\alpha K_\alpha, \bigoplus_\alpha \mu_{x_\alpha}\Big) = \mathrm{L}^2(\Omega, \mu).$$

Each K_α is compact and open in Ω, whence Ω is locally compact. On $Z(x_\alpha)$ the operator T acts as multiplication by z on $\mathrm{L}^2(K, \mu_{x_\alpha})$, and hence T acts on H as multiplication on $\mathrm{L}^2(\Omega, \mu)$ by a function m simply given by $m(z, \alpha) = z$ on K_α. This is a continuous function on Ω. □

if it is not invertible). Thus for any measure-preserving transformation T, the associated operator U_T is an isometry, and if T is invertible then the associated operator U_T is a unitary operator, called the *associated unitary operator* of T or *Koopman operator* of T.

A property of a measure-preserving transformation is said to be a *spectral* or *unitary property* if it can be detected by studying the associated operator on L^2_μ.

Lemma 2.18. *A measure-preserving transformation T is ergodic if and only if 1 is a simple eigenvalue of the associated operator U_T. Hence ergodicity is a unitary property.*

Proof. This follows from the proof of the equivalence of (2) and (5) in Proposition 2.14 or via Exercise 2.3.5 applied with $p = 2$: an eigenfunction for the eigenvalue 1 is a T-invariant function, and ergodicity is characterized by the property that the only T-invariant functions are the constants. □

Example 3: Lebesgue spectrum. Consider the unitary operator U on $\ell^2(\mathbb{Z})$ defined by the left shift:

$$U(a_i) = (a_{i-1}).$$

Clearly this operator has no eigenvectors. What is its spectral measure?

Theorem 2.5 *The shift operator on $\ell^2(\mathbb{Z})$ has Lebesgue spectrum of multiplicity one.*

(This means its spectral measure (class) μ is given by Lebesgue measure $|d\lambda|$ on the unit circle S^1.)

Proof. Observe that the map

$$(a_i) \mapsto \sum a_i z^i$$

gives an isomorphism between $\ell^2(\mathbb{Z})$ and $L^2(S^1)$. Since

$$zf(z) = z \sum a_i z_i = \sum a_{i-1} z^i,$$

this isomorphism sends U to the operator $L(f) = zf(z)$ acting on $L^2(S^1)$. Consequently the spectral measure of U is given simply by Lebesgue measure on S^1, with multiplicity one. ∎

spectrum of circle rotations

We have the unitary map $R^{\theta *}$ on $L^2(S^1)$ such that

$$R^{\theta *}(f)(z) := f(R^\theta(x))$$

On the Fourier basis $(e_k)_{k\in\mathbb{Z}}$ this acts by

$$R^{\theta *}(e_k) = \exp(2\pi i\theta)^k e_k$$

so it is diagonal with eigenvalues A_θ.

- If θ is rational, then its spectral measure is

$$\mu = \sum_{x\in A_\theta} \delta_x$$

  with multiplicity $M\Big|_{A_\theta} \equiv \infty$.
- If θ is irrational then its spectral measure is

$$\mu = \sum_{x\in A_\theta} \delta_x$$

  with multiplicity $M\Big|_{A_\theta} \equiv 1$

An irrational rotation of the circle has *pure point spectrum of multiplicity 1* supported on $\langle \lambda \rangle < S^1$ for some $\lambda \in S^1$ of infinite order that is either the rotation angle or its inverse. Thus, two irrational rotations by $\lambda_1, \lambda_2 \in S^1$ are isomorphic $\iff$ $\lambda_1 = \lambda_2^{\pm}$.

Haar measure on locally compact groups

locally compact Hausdorff spaces

Proposition: A closed subset of a compact space is compact.

If $K \subset X$ be closed where X is compact. If open sets in X

$$\{U_\alpha\}$$

(whose restriction to K) covers K then

$$\{U_\alpha\} \cup \{K^c\}$$

is an open cover of X. Thus it has a finite subcover, and thus we obtain a finite subcollection of an arbitrary cover $\{U_\alpha \cap K\}$, thus K is compact.

Proposition: If $K(\text{cpt}) \subset X$ for Hausdorff space X and $x \notin K$ then there are disjoint open subsets U, V of X such that $x \in U, K \subset V$ *(open sets separate compact sets and a point)*.

For each $y \in K$ choose disjoint open U_y, V_y where

$$x \in U_y, y \in V_y$$

- Then

$$\{V_y \mid y \in K\}$$

is an open cover of K so it has a finite subcover for $y_1, \ldots, y_n$. The open sets

$$U := \bigcap_j U_{y_j}, V := \bigcup_j V_{y_j}$$

have the desired properties.

Proposition: Every compact subset of a Hausdorff space is closed.

Proposition: Every compact Hausdorff space is normal.

(Urysohn's lemma) Let X be a normal space. If A, B are disjoint closed sets in X then there exists $f \in \mathcal{C}(X, [0,1])$ such that

$$f|_A \equiv 0, f|_B \equiv 1$$

Proposition: If X is locally compact Hausdorff and

$$x \in U(\text{open}) \subset X$$

then there is a compact $N \subset X$ such that

$$x \in N \subset U \subset X$$

If $\overline{U}$ is not compact, replace U by $U \cap K_x^\circ$ where K_x is a compact neighborhood of x, so we assume $\overline{U}$ is **compact**. Then ∂U is compact.

- By

> **Proposition:** If $K(\text{cpt}) \subset X$ for Hausdorff space X and $x \notin K$ then there are disjoint open subsets U, V of X such that $x \in U, K \subset V$ *(open sets separate compact sets and a point)*.

for $x, \partial U$ we have open sets V, W in $\bar{U}$ such that

$$x \in V, \partial U \subset W$$

- Then
  - V is open in X since $V \subset U$ is open.
  - $\bar{V}$ is closed in $\bar{U}$ so it is compact.
  - $\bar{V} \subset U \setminus W$
- So $N := \bar{V}$

> **Proposition:** If X is locally compact Hausdorff and
>
> $$K(\text{cpt}) \subset U(\text{open}) \subset X$$
>
> there exists a pre-compact open $V \subset X$ such that
>
> $$K \subset V \subset \underbrace{\bar{V}}_{\text{cpt}} \subset U \subset X$$

By

> **Proposition:** If X is locally compact Hausdorff and
>
> $$x \in U(\text{open}) \subset X$$
>
> then there is a compact $N \subset X$ such that
>
> $$x \in N \subset U \subset X$$

for each $x \in K$ we choose a compact

$$x \in N_x(\text{cpt}) \subset U$$

then

$$\{N_x^\circ\} \text{ cover } K$$

so there is a finite subcover

$$N_{x_1}^\circ, \ldots, N_{x_n}^\circ$$

then

$$K \subset V \subset \bar{V} := \bigcup_j N_{x_j}$$

is compact and

$$\subset U$$

> **Proposition: (local Urysogn's lemma)** Let X be locally compact and
>
> $$K(\text{cpt}) \subset U(\text{open}) \subset X$$
>
> Then there exists
>
> $$f : X \to [0, 1]$$

continuous with compact support inside U and

$$f|_K \equiv 1$$

Let $\bar{V}$ be as in

> **Proposition:** If X is locally compact Hausdorff and
>
> $$K(\text{cpt}) \subset U(\text{open}) \subset X$$
>
> there exists a pre-compact open $V \subset X$ such that
>
> $$K \subset V \subset \underbrace{\bar{V}}_{\text{cpt}} \subset U \subset X$$

- As $\bar{V}$ is compact, it is normal

> **Proposition:** Every compact Hausdorff space is normal.

- Then by

> **(Urysohn's lemma) Let X be a normal space. If A, B are disjoint closed sets in X then there exists $f \in \mathcal{C}(X, [0,1])$ such that**
>
> $$f|_A \equiv 0, f|_B \equiv 1$$

  there exists

$$f \in \mathcal{C}(\bar{V}, [0,1])$$

  such that

$$f|_K \equiv 1, f|_{\partial V} \equiv 0$$

- We extend f to X by setting $f \equiv 0$ on $\bar{V}^c$ by pasting lemma.
- We observe

$$\text{supp}(\phi) \subset \bar{V} \subset U$$

Locally compact groups

group	invariant measure
discrete group (finite, $\mathbb{Z}$, etc) $\iff$ countable (if locally cpt)	counting measure
$(\mathbb{R}, +)$	Lebesgue measure
$(\mathbb{R}_{>0}, \times)$	$$\frac{\mathrm{d}x}{x}$$
$(\mathbb{Q}, +)$ (countable but not discrete)	no Haar measure!
$SU(3)$	measure on S^3
$GL(n, \mathbb{R})$	$$\frac{\mathrm{d}m(g)}{\lVert\det g\rVert^n}$$

Any nonempty countable complete metric space has an isolated point (if it didn't, then each singleton would have empty interior but the union of countably many singletons is the whole space, violating the Baire category theorem). If this space is a topological group, this then implies it is discrete. [1]

Hilbert's fifth problem [edit]

There are several strong results on the relation between topological groups and Lie groups. First, every continuous homomorphism of Lie groups $G \to H$ is smooth. It follows that a topological group has a unique structure of a Lie group if one exists. Also, Cartan's theorem says that every closed subgroup of a Lie group is a Lie subgroup, in particular a smooth submanifold.

Hilbert's fifth problem asked whether a topological group G that is a topological manifold must be a Lie group. In other words, does G have the structure of a smooth manifold, making the group operations smooth? As shown by Andrew Gleason, Deane Montgomery, and Leo Zippin, the answer to this problem is yes.[13] In fact, G has a real analytic structure. Using the smooth structure, one can define the Lie algebra of G, an object of linear algebra that determines a connected group G up to covering spaces. As a result, the solution to Hilbert's fifth problem reduces the classification of topological groups that are topological manifolds to an algebraic problem, albeit a complicated problem in general.

The theorem also has consequences for broader classes of topological groups. First, every compact group (understood to be Hausdorff) is an inverse limit of compact Lie groups. (One important case is an inverse limit of finite groups, called a profinite group. For example, the group $\mathbb{Z}_p$ of p-adic integers and the absolute Galois group of a field are profinite groups.) Furthermore, every connected locally compact group is an inverse limit of connected Lie groups.[14] At the other extreme, a totally disconnected locally compact group always contains a compact open subgroup, which is necessarily a profinite group.[15] (For example, the locally compact group $\mathrm{GL}(n,\mathbb{Q}_p)$ contains the compact open subgroup $\mathrm{GL}(n,\mathbb{Z}_p)$, which is the inverse limit of the finite groups $\mathrm{GL}(n,\mathbb{Z}/p^r)$ as r' goes to infinity.)

Haar measure

A Borel measure is called **left invariant** if

$$x \in G, E \in \mathfrak{B}_G \implies \mu(xE) = \mu(E)$$

A linear functional

$$I : \mathcal{C}_c(G) \to \mathbb{C}?$$

is **left invariant** if

$$x \in G \implies I(L_x f) = I(f)$$

$$(L_y f)(x) := f(y^{-1}x)$$

$$\mathcal{C}_c^+(G) := \{f \in \mathcal{C}_c(G) \mid f \geq 0, \|f\|_\infty > 0\}$$

A **topological group** is a group G endowed with a topology such that the group operations $(x, y) \mapsto xy$ and $x \mapsto x^{-1}$ are continuous from $G \times G$ and G to G. Examples include topological vector spaces (the group operation being addition), groups of invertible $n \times n$ real matrices (with the relative topology induced from $\mathbb{R}^{n\times n}$), and all groups equipped with the discrete topology. If G is a topological group, we denote the identity element of G by e, and for $A, B \subset G$ and $x \in G$ we define

$$xA = \{xy : y \in A\}, \qquad Ax = \{yx : y \in A\},$$
$$A^{-1} = \{x^{-1} : x \in A\}, \qquad AB = \{yz : y \in A,\ z \in B\}.$$

We say that $A \subset G$ is **symmetric** if $A = A^{-1}$.

Here are some of the basic properties of topological groups:

11.1 Proposition. *Let G be a topological group.*

a. *The topology of G is translation invariant: If U is open and $x \in G$, then Ux and xU are open.*

b. *For every neighborhood U of e there is a symmetric neighborhood V of e with $V \subset U$.*
c. *For every neighborhood U of e there is a neighborhood V of e with $VV \subset U$.*
d. *If H is a subgroup of G, so is $\overline{H}$.*
e. *Every open subgroup of G is also closed.*
f. *If K_1, K_2 are compact subsets of G, so is K_1K_2.*

Proof. (a) is equivalent to the continuity in each variable of the map $(x, y) \mapsto xy$, and (b) and (c) are equivalent to the continuity of $x \mapsto x^{-1}$ and $(x, y) \mapsto xy$ at the identity. (Details are left to the reader.) For (d), if $x, y \in \overline{H}$, there exist nets $\langle x_\alpha \rangle_{\alpha\in A}$, $\langle y_\beta \rangle_{\beta\in B}$ in H that converge to x and y. Then $x_\alpha^{-1} \to x^{-1}$ and $x_\alpha y_\beta \to xy$ (with the usual product ordering on $A \times B$), so x^{-1} and xy belong to $\overline{H}$. For (e), if H is an open subgroup, the cosets xH are open for all x, so that $G \setminus H = \bigcup_{x\notin H} xH$ is open and hence H is closed. Finally, (f) is true because K_1K_2 is the image of the compact set $K_1 \times K_2$ under the continuous map $(x, y) \mapsto xy$. ∎

If f is a continuous function on the topological group G and $y \in G$, we define the left and right translates of f through y by

$$L_y f(x) = f(y^{-1}x), \qquad R_y f(x) = f(xy).$$

(The point of using y^{-1} on the left and y on the right is to make $L_{yz} = L_y L_z$ and $R_{yz} = R_y R_z$.) f is called **left** (resp. **right**) **uniformly continuous** if for every $\epsilon > 0$ there is a neighborhood V of e such that $\|L_y f - f\|_u < \epsilon$ (resp. $\|R_y f - f\|_u < \epsilon$) for $y \in V$. (Some authors reverse the roles of L_y and R_y in this definition.)

11.2 Proposition. *If $f \in C_c(G)$, then f is left and right uniformly continuous.*

Proof. We shall consider right uniform continuity; the proof on the left is the same. Let $K = \operatorname{supp}(f)$ and suppose $\epsilon > 0$. For each $x \in K$ there is a neighborhood U_x of e such that $|f(xy) - f(x)| < \frac{1}{2}\epsilon$ for $y \in U_x$, and by Proposition 11.1(b,c) there is a symmetric neighborhood V_x of e with $V_x V_x \subset U_x$. Then $\{xV_x\}_{x \in K}$ covers K, so there exist $x_1, \ldots, x_n \in K$ such that $K \subset \bigcup_1^n x_j V_{x_j}$. Let $V = \bigcap_1^n V_{x_j}$; we claim that $|f(xy) - f(x)| < \epsilon$ if $y \in V$. On the one hand, if $x \in K$, then for some j we have $x_j^{-1}x \in V_{x_j}$ and hence $xy = x_j(x_j^{-1}x)y \in x_j U_{x_j}$; therefore,

$$|f(xy) - f(x)| \le |f(xy) - f(x_j)| + |f(x_j) - f(x)| < \epsilon.$$

On the other hand, if $x \notin K$, then $f(x) = 0$, and either $f(xy) = 0$ (if $xy \notin K$) or $x_j^{-1}xy \in V_{x_j}$ for some j (if $xy \in K$); in the latter case $x_j^{-1}x = x_j^{-1}xyy^{-1} \in U_{x_j}$, so that $|f(x_j)| < \frac{1}{2}\epsilon$ and hence $|f(xy)| < \epsilon$. ∎

If $E \in \mathfrak{B}_G$ and V is open, non-empty, then the smallest of number of left translates of V that cover E

Definition.

$$(E:V) := \inf\{|A| \mid E \subset \bigcup_{x \in A} xV\} \subset \{1, 2, \ldots, \infty\}$$

gives a rough estimate of the relative sizes of E and V.

Fixing a pre-compact open set E_0 the ratio

$$E \mapsto \frac{(E:V)}{(E_0:V)}$$

gives a rough estimate of size of E relative to E_0.

Shifting to $\mathcal{C}_c(G)$ side: if $f, \phi \in \mathcal{C}_c^+(G)$ then

$$V(\phi) := \left\{x \mid \phi(x) > \frac{1}{2}\|\phi\|_\infty\right\}$$

is **open and non-empty** (as $\|\phi\|_\infty > 0$), so finitely many translates of it cover $\operatorname{supp} f$

$$\operatorname{supp} f \subset \bigcup_{j=1}^{n} x_j V(\phi)$$

As

$$\begin{aligned} yV(\phi) &= \left\{ yx \mid \phi(x) > \frac{\|\phi\|_\infty}{2} \right\} \\ &= \left\{ z \mid \phi(y^{-1}z) = L_y\phi(z) > \frac{\|\phi\|_\infty}{2} \right\} \\ &= V(L_y\phi) \end{aligned}$$

$$1 < \frac{2}{\|\phi\|_\infty} L_y\phi(z)$$

Then

$$\frac{f(z)}{\|f\|_\infty} \leq 1 < \frac{2}{\|\phi\|_\infty} L_y\phi(z)$$

for each $z \in \operatorname{supp} f$, for any $y \in G$.

Thus

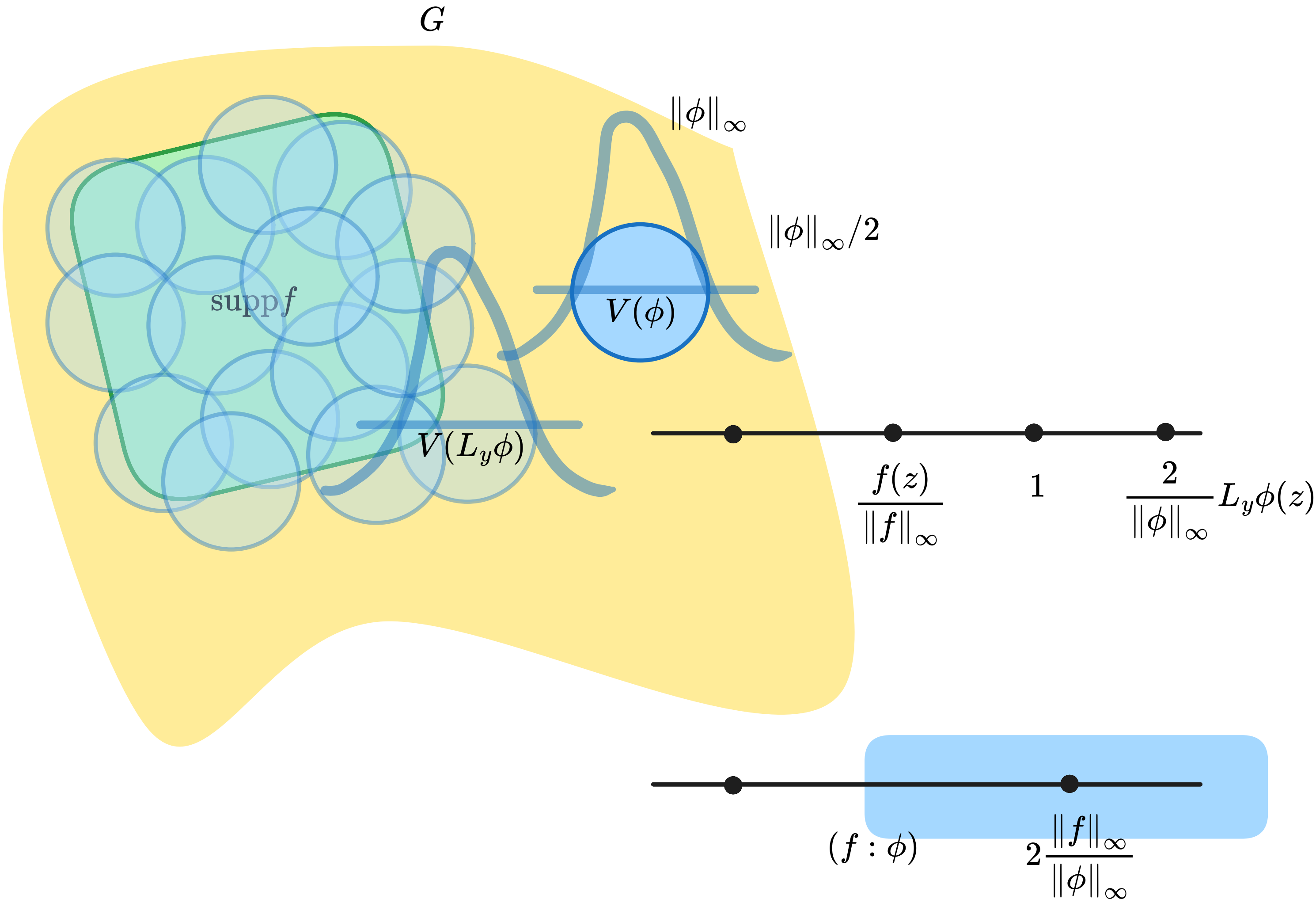
$$f \leq 1 < \frac{2}{n} \frac{\|f\|_\infty}{\|\phi\|_\infty} \sum_{j=1}^{n} L_{x_j}\phi$$

We interpret this as

$$f \leq \sum_{j=1}^{n} \left(\frac{2}{n} \frac{\|f\|_\infty}{\|\phi\|_\infty} \right) L_{x_j}\phi$$

Now the "Haar covering number"

$$(f : \phi) := \inf \left\{ \sum_{i=1}^{n} c_j \middle| f \leq \sum_{j=1}^{n} c_j L_{x_j}\phi \begin{array}{r} \text{for some } n \in \mathbb{N} \\ \text{and } x_1, \ldots, x_n \in G \end{array} \right\}$$



and

$$0 < \frac{\|f\|_\infty}{\|\phi\|_\infty} \leq (f:\phi) \leq 2\frac{\|f\|_\infty}{\|\phi\|_\infty}$$

Proposition:

- $(f:\phi) = (L_x f : \phi)$ for any $x \in G$
- $(cf:\phi) = c(f:\phi)$ for any $c > 0$
- $(f+g:\phi) \leq (f:\phi) + (g:\phi)$
- $(f:\phi) \leq (f:g)(g:\phi)$

We have

$$f \leq \sum_j c_j L_{x_j}\phi \iff L_x f \leq \sum_j c_j L_{xx_j}\phi$$

- $$cf \leq \sum_j c_j L_{x_j}\phi \iff f \leq \frac{1}{c}\sum_j c_j L_{x_j}\phi$$
- Given $f \leq \sum_j c_j L_{x_j}\phi, g \leq \sum_j d_j L_{y_j}\phi$ then
$$f + g \leq \sum_j c_j L_{x_j}\phi + \sum_j d_j L_{x_j}\phi$$
- Given $f \leq \sum_j c_j L_{x_j} g, g \leq \sum_j d_j L_{y_j}\phi$ then
$$f \leq \sum_{j,k} c_j d_k L_{x_j y_k}\phi$$

Definition. Fix $f_0 \in \mathcal{C}_c^+(G)$ then

$$\begin{aligned} I_\phi : \mathcal{C}_c^+(G) &\to [0, \infty) \\ f &\mapsto \frac{(f:\phi)}{(f_0:\phi)} \end{aligned}$$

It is

- left-invariant
- $$I_\phi(cf) = cI_\phi(f)$$
- sub-additive
  $$I_\phi(f_1 + f_2) \leq I_\phi(f_1) + I_\phi(f_2)$$
- uniformly "bounded" with respect to ϕ
  $$I_\phi(f) \in \left[\frac{1}{(f_0:f)}, (f:f_0)\right]$$

Proposition: If $f_1, f_2 \in \mathcal{C}_c^+(G)$ and $\epsilon > 0$ there is a neighborhood V_ϵ of i_G such that

$$I_\phi(f_1) + I_\phi(f_2) \leq I_\phi(f_1 + f_2) + \epsilon$$

for every $\phi \in \mathcal{C}_c^+(G)$ whose $\operatorname{supp}(\phi) \subset V_\epsilon$.

- Fix $g \in \mathcal{C}_c^+(G)$ such that $g = 1$ on $\operatorname{supp}(f_1 + f_2)$ and $\delta > 0$. Set
  $$\begin{aligned} h &:= f_1 + f_2 + \delta g \\ h_i &:= f_i / h \end{aligned}$$
- There is a neighborhood V_δ of i_G such that
  $$y^{-1}x \in V \implies |h_i(x) - h_i(y)| < \delta \text{ for } i = 1, 2$$
- Choose $\phi \in \mathcal{C}_c^+(G)$ with $\operatorname{supp}(\phi) \subset V$ and
  $$h \leq \sum_j c_j L_{x_j}\phi$$

  then

  $$x_j^{-1}x \in \operatorname{supp}(\phi) \implies |h_i(x) - h_i(x_j)| < \delta$$
- So
  $$\begin{aligned} f_i(x) &= h(x)h_i(x) \\ &\leq \sum_j c_j \phi(x_j^{-1}x) h_i(x) \\ &\leq \sum_j c_j (h_i(x_j) + \delta) L_{x_j}\phi(x) \end{aligned}$$

$$\implies (f_i : \phi) \leq \sum_j c_j(h_i(x_j) + \delta)$$

$$(f_1 : \phi) + (f_2 : \phi) \leq \sum_j c_j(1 + 2\delta)$$

- $$\begin{aligned} I_\phi(f_1) + I_\phi(f_2) &\leq (1 + 2\delta) I_\phi(h) \\ &\leq (1 + 2\delta)(I_\phi(f_1 + f_2) + \delta I_\phi(g)) \end{aligned}$$
- Choose δ small enough such that

$$\begin{gathered} (1 + 2\delta)(I_\phi(f_1 + f_2) + \delta I_\phi(g)) \\ \leq 2\delta(f_1 + f_2 : f_0) + \delta(1 + 2\delta)(g : f_0) < \epsilon \end{gathered}$$

Every locally compact group G posses a left Haar measure and it is unique up to a multiplicative constant.

For each $f \in \mathcal{C}_c^+(G)$ let

$$X_f := \left[\frac{1}{(f_0 : f)}, (f : f_0) \right]$$

and let

$$X := \prod_{f \in \mathcal{C}_c^+(G)} X_f$$

which is a **compact Hausdorff** space by *Tychonoff's theorem*.

Then for every $\phi \in \mathcal{C}_c^+(G)$

$$I_\phi : \mathcal{C}_c^+(G) \to \mathbb{R}_{>0}$$

is an element of X, that is

$$\{I_\phi \mid \phi \in \mathcal{C}_c^+(G)\} \subset X$$

Intuition

Because of

> **Proposition:** If $f_1, f_2 \in \mathcal{C}_c^+(G)$ and $\epsilon > 0$ there is a neighborhood V_ϵ of i_G such that
>
> $$I_\phi(f_1) + I_\phi(f_2) \leq I_\phi(f_1 + f_2) + \epsilon$$
>
> for every $\phi \in \mathcal{C}_c^+(G)$ whose $\mathrm{supp}(\phi) \subset V_\epsilon$.

to find the desired *linear functional* $I \in X$ we need to somehow "take the limit" I_ϕ as $\mathrm{supp}(\phi) \to \emptyset$.

- For every compact $V \subset G$ containing i_G let

$$K(V) := \overline{\{I_\phi \mid \mathrm{supp}(\phi) \subset V\}} \subset X$$

- So

$$\{K(V) \mid i_G \in V(\mathrm{cpt}) \subset G\}$$

is a collection of closed sets in the compact space X.

- Clearly,

$$K(\cap_j V_j) \subset \cap_j K(V_j)$$

so this collection has the *finite intersection property*.

Hence, by

A topological space is compact $\iff$ for every family $\{F_\alpha \mid \alpha \in A\}$ of closed sets with *finite intersection property* ($\cap_{\beta \in B} F_\beta \neq \emptyset$ for all finite $B \subset A$) the intersection is non-empty

$$\bigcap_{\alpha \in A} F_\alpha \neq \emptyset$$

Let $U_\alpha := F_\alpha^c$ be the corresponding collection of open sets. Then

- U_α has finite intersection property

$$\cap_{\beta \in B} F_\beta \neq \emptyset \text{ for all finite } B \subset A$$

$\iff$ there is no subfamily of $\{U_\alpha\}$ that covers X

$$\cup_{\beta \in B} U_\beta \neq X \text{ for all finite } B \subset A$$

- and

$$\bigcap_{\alpha \in A} F_\alpha \neq \emptyset \iff \{U_\alpha \mid \alpha \in A\} \text{ cover } X$$

there is an element I in the intersection of all of $K(V)$.

- For any V, every neighborhood of I in X intersects $\{I_\phi \mid \mathrm{supp}(\phi) \subset V\}$.
- That is, for any compact neighborhood V of i_G, and any $f_1, \dots, f_n \in \mathcal{C}_c^+(G)$ and $\epsilon > 0$ there exists $\phi \in \mathcal{C}_c^+(G)$ with $\mathrm{supp}(\phi) \subset V$ such that

$$|I(f_j) - I_\phi(f_j)| < \epsilon$$

- By

  Proposition: If $f_1, f_2 \in \mathcal{C}_c^+(G)$ and $\epsilon > 0$ there is a neighborhood V_ϵ of i_G such that

  $$I_\phi(f_1) + I_\phi(f_2) \leq I_\phi(f_1 + f_2) + \epsilon$$

  for every $\phi \in \mathcal{C}_c^+(G)$ whose $\mathrm{supp}(\phi) \subset V_\epsilon$.

  $$I(f_1) + I(f_2) - (I_\phi(f_1) + I_\phi(f_2)) \leq \epsilon$$

  and

  $$I_\phi(f_1 + f_2) - I(f_1 + f_2) \leq \epsilon$$

  we get

  $$I(f_1) + I(f_2) - I(f_1 + f_2) \leq 3\epsilon$$

  for all $\epsilon > 0$. Hence

  $$I$$

  is linear.

Thus by

positive measures and positive functions on $\mathcal{C}_c$

A finite positive Borel measure μ on X gives a functional

$$\Lambda : \mathcal{C}_c(X) \to k$$
$$\Lambda(f) := \int_{(X,\mu)} f$$

Thus functional is *positive* in the sense that if $f(X) \subset (0,1)$ then $(\Lambda f)(X) \subset (0,1)$.

[1]

The converse of this statement is true!

(Riesz representation of positive functionals on $\mathcal{C}_c$) Let X be locally compact Hausdorff space and Λ be a positive linear functional on $\mathcal{C}_c(X)$. Then there exists a σ-algebra in X which contains all Borel sets in X and there exists a unique positive Radon measure μ in this σ-algebra such that

$$\Lambda(f) = \int_{(X,\mu)} f$$

for every $f \in \mathcal{C}_c(X)$ and for every compact $K \subseteq X$

$$\mu(K) < \infty$$

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1. math.uchicago.edu/~may/REU2023/REUPapers/Espejo.pdf#page=9.37 ↩

1. https://math.stackexchange.com/a/2770136/1290493 ↩

Actions of locally compact groups

Definition. Ergodic group actions

A G-action is **ergodic** if any measurable $A \in \mathfrak{B}_G$

$$\forall g \in G : g^{-1}A = A \implies \mu(A) = 0 \text{ or } 1$$

Proposition: Let G be a σ-locally compact metrizable group with left Haar measure ν_G. Then for any two Borel sets $B_1, B_2 \in \mathfrak{B}_G$ with positive measure $\nu_G(B_1)\nu_G(B_2) > 0$, the sets

$$\{g \in G | \nu_G(gB_1 \cap B_2) > 0\}$$
$$\{g \in G | \nu_G(B_1 g \cap B_2) > 0\}$$

are non-empty open in G. Moreover

$$\nu_G(V) > 0 \iff \nu_G(B^{-1})$$

for $B \in \mathfrak{B}_G$.

$SL(2,\mathbb{R}) \curvearrowright \mathbb{R}H^2$

#Wiki/group-action/Lie

- We observe

$$\Im\left(\frac{az+b}{cz+d}\right) = \frac{\Im(z)}{|cz+d|^2}$$

  for $a,b,c,d \in \mathbb{R}$

Definition. $SL(2,\mathbb{R}) \curvearrowright \mathbb{R}H^2$

The group $SL(2,\mathbb{R})$ acts on the upper-half plane $\mathbb{R}H^2$ by *Mobius transformations*

$$(x,y) \overset{\begin{bmatrix} a & b \\ c & d \end{bmatrix}}{\longmapsto} \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}\right)^{\mathsf{T}}$$

for $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in SL(2,\mathbb{R})$ and $(x,y) \in \mathbb{R}H^2$, where the action by $PSL(2,\mathbb{R})$ is faithful.

> group of complex 1-manifold automorphisms

- Let $x+iy \in H^2$. Then

$$\begin{bmatrix} \sqrt{y} & x/\sqrt{y} \\ 0 & 1/\sqrt{y} \end{bmatrix}$$

  maps

$$i \mapsto \frac{\sqrt{y}\,i + \frac{x}{\sqrt{y}}}{\frac{1}{\sqrt{y}}} = x+iy$$

  - Thus the action $SL(2,\mathbb{R}) \curvearrowright \mathbb{R}H^2$ is **transitive**.
- Now, the stabilizer of $i \in \mathbb{R}H^2$ must have

$$|ci+d|^2 = 1$$

  by

  > - We observe
  >
  > $$\Im\left(\frac{az+b}{cz+d}\right) = \frac{\Im(z)}{|cz+d|^2}$$
  >
  > for $a,b,c,d \in \mathbb{R}$

  - That means $(c,d) = (\sin\theta, \cos\theta)$ for some $\theta \in \mathbb{R}$, but

$$\begin{aligned} \frac{ai+b}{\sin\theta\, i + \cos\theta} &= i \\ \implies a,b &= \cos\theta, -\sin\theta \end{aligned}$$

  - Thus

$$\mathrm{stab}_{SL(2,\mathbb{R})}(i) = SO(2,\mathbb{R})$$

The transitive, faithful action $PSL(2,\mathbb{R}) \curvearrowright \mathbb{R}H^2$ by *Mobius transformations* is the connected component of isometry group of the hyperbolic metric on upper-half plane

$$\mathrm{AutMan}_O(\mathbb{R}H^2, g_{\mathrm{Hy}})_0 \cong PSL(2,\mathbb{R})$$

Therefore the action is symplectic or area preserving in the hyperbolic area 2-form.

Thus, we have

$$SL(2,\mathbb{R}) \curvearrowright (\mathbb{R}H^2, g_{\mathrm{Hy}}, \lambda_{\mathrm{Hy}})$$

> Lie generators

For the transitive action $SL(2,\mathbb{R}) \curvearrowright \mathbb{R}H^2$ by *Mobius transformations*, the stabilizer of $i = (0,1) \in H$ is $SO(2,\mathbb{R}) < SL(2,\mathbb{R})$. Thus

$$\frac{SL(2,\mathbb{R})}{SO(2,\mathbb{R})} \cong_{\mathsf{Man}} \mathbb{R}H^2$$

by orbit-stabilizer theorem.

sub-actions

- how subgroups act
    - $SL(2,\mathbb{Z}) \curvearrowright \mathbb{R}H^2$
    - Γ torsion-free discrete
    - ...

induced action on $T^{\mathsf{u}}H^2$

$$A(z) = \frac{az+b}{cz+d}$$

then

- its *complex* derivative is

$$\begin{aligned} A'(z) &= \frac{a(cz+d) - c(az+b)}{(cz+d)^2} \\ &= \frac{\det A}{(cz+d)^2} \\ &= \frac{1}{(cz+d)^2} \end{aligned}$$

- which is precisely the multiplier for the linear map $\mathfrak{D}_z A$, making the the derivative

$$\begin{aligned} \mathfrak{D}A : T\mathbb{R}H^2 &\to T\mathbb{R}H^2 \\ (z,v) &\mapsto \left(\frac{az+b}{cz+d}, \frac{v}{(cz+d)^2}\right) \end{aligned}$$

  where $T\mathbb{R}H^2 \cong \mathbb{R}H^2 \times U(1)$ naturally.
- The action preserves the hyperbolic metric because

$$\begin{aligned} \langle D_z A(v), D_z A(w)\rangle_{\mathrm{Hyp}} &= \left(\frac{y}{|cz+d|^2}\right)^{-2} \left(\frac{v}{(cz+d)^2}, \frac{w}{(cz+d)^2}\right) \\ &= \frac{1}{y^2}\langle v, w\rangle_{\mathrm{DOT}} \\ &= \langle v, w\rangle_{\mathrm{Hyp}} \end{aligned}$$

- As the inner product is preserved, hyperbolic unit vectors map to themselves, thus we restrict the action to

$$SL(2,\mathbb{R}) \curvearrowright T^{\mathrm{u}}\mathbb{R}H^2$$

- Now consider the action of the stabilizers

$$A = \frac{i\cos\theta - \sin\theta}{i\sin\theta + \cos\theta}$$

of $i \in \mathbb{R}H^2$

$$(D_i A)(v) = \frac{v}{(i\sin\theta + \cos\theta)^2} = (\cos 2\theta - i\sin 2\theta)v$$

  - So varying θ, we can map v to any element of $T^{\mathrm{u}}_i\mathbb{R}H^2$.
  - Thus, the action

$$SL(2,\mathbb{R}) \curvearrowright T^{\mathrm{u}}\mathbb{R}H^2$$

  - Now, say A is a stabilizer of (i, v), then

$$\begin{aligned} 2\theta \bmod 2\pi &= 0 \\ \implies \theta &= \pi\mathbb{Z} \\ \implies A &= \pm I \end{aligned}$$

  - Thus, the action

$$PSL(2,\mathbb{R}) \curvearrowright T^{\mathrm{u}}\mathbb{R}H^2$$

  is transitive and free giving us an isomorphism

$$\begin{aligned} PSL(2,\mathbb{R}) &\cong T^{\mathrm{u}}\mathbb{R}H^2 \\ A &\mapsto A(i,i) \end{aligned}$$

Definition. $SL(2,\mathbb{R}) \curvearrowright T\mathbb{R}H^2 \cong \mathbb{R}H^2 \times U(1)$

We have the action

$$\begin{aligned} SL(2,\mathbb{R}) &\curvearrowright T\mathbb{R}H^2 \cong \mathbb{R}H^2 \times U(1) \\ (z,v) &\xrightarrow{\begin{bmatrix} a & b \\ c & d \end{bmatrix}} \left(\frac{az+b}{cz+d}, \frac{v}{(cz+d)^2} \right) \end{aligned}$$

induced by $SL(2,\mathbb{R}) \curvearrowright \mathbb{R}H^2$ where the action by $PSL(2,\mathbb{R})$ is free and transitive giving us an isomorphism of left $PSL(2,\mathbb{R})$-manifolds

$$\begin{aligned} PSL(2,\mathbb{R}) &\cong T^{\mathrm{u}}\mathbb{R}H^2 \\ A &\mapsto A(i,i) \end{aligned}$$

geodesic flow

We have the geodesic flow

$$\exp(t-) : T^{\mathrm{u}}\mathbb{R}H^2 \to T^{\mathrm{u}}\mathbb{R}H^2$$

of the hyperbolic metric on $\mathbb{R}H^2$

- We have

$$\exp(t(i,i)) = (e^t i, e^t i) = \mathfrak{D}_i \begin{bmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{bmatrix} (i)$$

- For the geodesic starting at $(z, v) = A(i, i)$ we have

$$\begin{aligned} \exp(t(z,v)) &= \mathfrak{D}A(\exp(t(i,i))) \\ &= \mathfrak{D}A\left(\mathfrak{D}_i \begin{bmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{bmatrix} (i)\right) \\ &= \mathfrak{D}_i \left(A \begin{bmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{bmatrix}^{-1}\right)(i) \end{aligned}$$

- Thus, under the identification

Definition. $SL(2,\mathbb{R}) \curvearrowright T\mathbb{R}H^2 \cong \mathbb{R}H^2 \times U(1)$

We have the action

$$\begin{aligned} SL(2,\mathbb{R}) &\curvearrowright T\mathbb{R}H^2 \cong \mathbb{R}H^2 \times U(1) \\ (z,v) &\xrightarrow{\begin{bmatrix} a & b \\ c & d \end{bmatrix}} \left(\frac{az+b}{cz+d}, \frac{v}{(cz+d)^2}\right) \end{aligned}$$

induced by $SL(2,\mathbb{R}) \curvearrowright \mathbb{R}H^2$ where the action by $PSL(2,\mathbb{R})$ is free and transitive giving us an isomorphism of left $PSL(2,\mathbb{R})$-manifolds

$$\begin{aligned} PSL(2,\mathbb{R}) &\cong T^{u}\mathbb{R}H^2 \\ A &\mapsto A(i,i) \end{aligned}$$

the geodesic flow corresponds to right multiplication by $\begin{bmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{bmatrix}^{-1}$ that is

$$A \xrightarrow{t} A \begin{bmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{bmatrix}^{-1}$$

geodesic flow on quotients of PSL2

Let $\Gamma \leq PSL(2,\mathbb{R})$ be a lattice. Then any $g(t) := R_{a_t}$ for some $t \neq 0$ is an ergodic transformation on

$$\left(\frac{PSL(2,\mathbb{R})}{\Gamma}, \nu_X\right)$$

We fix a $t \neq 0$ and normalize the Haar measure ν_X to ensure $\nu_X(X) = 1$ and let

$$f : X \to \mathbb{R}$$

be a measurable $R_{a_t} =: R$-invariant function.

steps	roughly
Proposition: $$B := \left\{ x \middle\vert \lim_{n\to\infty} \frac{1}{n} \sum_{l=0}^{n-1} \chi_K(R^l x) > \frac{1}{2} \right\}$$ has measure $\nu_X(B) \geq 1 - 2\epsilon$	more than $\frac{1}{2}$ of future of $x \in B$ belongs to K
• Then because $$f(-) = f(R^l(-))$$ for x, y and for all $l \geq 1$ we have $$\begin{aligned} d_X(R^l(x), R^l(y)) &= d_X(xa_t^{-l}, xu^-(-s)a_t^l) \\ &\leq d_{PSL(2,\mathbb{R})}(I_2, a_t^l u^-(-s)a_t^{-l}) \xrightarrow{l\to\infty} 0 \end{aligned}$$	simultaneous times $l \geq 0$ with $R^l x, R^l y \in K$
	$$R^n(x) = R^n(x)a_t^n u^- a_t^{-n},\ R^n(x)$$ are close together for $n \gg 1$ so $$f(x) = f(R^n(x)) \approx f(R^n(y)) = f(y)$$

Proposition:

$$B := \left\{ x \middle\vert \lim_{n\to\infty} \frac{1}{n} \sum_{l=0}^{n-1} \chi_K(R^l x) > \frac{1}{2} \right\}$$

has measure $\nu_X(B) \geq 1 - 2\epsilon$

- By

  (Pointwise ergodic theorem for measure preserving transformation) Let $(X, \mathfrak{M}_X, \mu)$ be a probability space, $T : X \to X$ be μ-preserving and $f \in L^1(X,\mu)$. Then

  $$\langle f \rangle_T = \lim_{n\to\infty} A_T^{(n)}(f) = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} f \circ T^k$$

converges pointwise almost everywhere on X to a T-invariant function $\langle f\rangle_T \in L^1(X,\mu)^T$ and

$$\int_X \langle f\rangle_T \mathrm{d}\mu = \int_X f \mathrm{d}\mu$$

$$\begin{aligned}\chi_K^* : X &\to [0,1]\\ x &\mapsto \lim_{n\to\infty} \frac{1}{n}\sum_{l=0}^{n-1} \chi_K(R^l x)\end{aligned}$$

exists almost everywhere and

$$\int_X \chi_K^* \mathrm{d}\nu_X = \nu_X(K) \geq 1-\epsilon$$

- Thus,

$$\nu_X(B) \geq 1-2\epsilon$$

- So,

$$\begin{aligned}1-\epsilon \leq \int_B \chi_K^* \mathrm{d}\nu_X + \int_{X\setminus B} \chi_K^* \mathrm{d}\nu_X &\leq \nu_X(B) + \frac{1}{2}\nu_X(X\setminus B)\\ &= \nu_X(B) + \frac{1}{2}(1-\nu_X(B))\\ &= \frac{1}{2}\nu_X(B) + \frac{1}{2}\end{aligned}$$

- Thus

$$\nu_X(B) \geq 1-2\epsilon$$

- Let $x, y := xu^-(s) \in B$ for some $s \in \mathbb{R}$.
  - Then because

  $$f(-) = f(R^l(-))$$

  for x, y and for all $l \geq 1$ we have

  $$\begin{aligned}d_X(R^l(x), R^l(y)) &= d_X(xa_t^{-l}, xu^-(-s)a_t^l)\\ &\leq d_{PSL(2,\mathbb{R})}(I_2, a_t^l u^-(-s)a_t^{-l}) \xrightarrow{l\to\infty} 0\end{aligned}$$

  - By

  Proposition:

  $$B := \left\{x \middle| \lim_{n\to\infty} \frac{1}{n}\sum_{l=0}^{n-1} \chi_K(R^l x) > \frac{1}{2}\right\}$$

  has measure $\nu_X(B) \geq 1-2\epsilon$

  we have a common sequence $l_n \xrightarrow{n\to\infty} \infty$ such that

  $$R^{l_n}(x), R^{l_n}(y) \in K$$

-

Coifman and Weiss - Transference Methods in Analysis (1977)

Definition. Hilbert transform on $\mathbb{R}$

For $p \in [0, \infty]$, the **Hilbert transform** is the operator

$$\begin{aligned} H : L^p(\mathbb{R}) &\to L^p(\mathbb{R}) \\ f \mapsto Hf &:= \lim_{\epsilon \to 0+} \underbrace{\frac{1}{\pi} \int_{\epsilon \leq |t|} \frac{f(s-t)}{t}}_{H_\epsilon f} \end{aligned}$$

- for $p = 1$, this operator is weak-L^1 bounded
- for $p > 1$, this operator is bounded

Proposition: For the Hilbert transform

Definition. Hilbert transform on $\mathbb{R}$

For $p \in [0, \infty]$, the **Hilbert transform** is the operator

$$\begin{aligned} H : L^p(\mathbb{R}) &\to L^p(\mathbb{R}) \\ f \mapsto Hf &:= \lim_{\epsilon \to 0+} \underbrace{\frac{1}{\pi} \int_{\epsilon \leq |t|} \frac{f(s-t)}{t}}_{H_\epsilon f} \end{aligned}$$

if $1 < p < \infty$ we have

$$f \in L^p(\mathbb{R}) \implies \|H_\epsilon f\|_p \leq A_p \|f\|_p$$

singular integral transforms

Let $h \in L^1(\mathbb{R}^n)$. Then

$$\int_{\mathbb{R}^n} h = \int_{\hat{x} \in S^{n-1}} \left(\int_{r \in (0,\infty)} h(r\hat{x}) r^{n-1} \right)$$

Definition. Singular integral transform in $\mathbb{R}^n$ with homogeneous odd kernels

Let

$$k(y) := \frac{\Omega(\hat{y})}{|y|^n}$$

where $\Omega \in L^1(S^{n-1})$ and $\Omega(-\hat{y}) = -\Omega(\hat{y})$.

Then

$$
\begin{aligned}
(H_k^\epsilon f)(x) &:= \int_{\epsilon\le|y|\le\frac{1}{\epsilon}} k(y)f(x-y) \\
&= \int_{\hat{y}\in S^{n-1}} \Omega(\hat{y}) \int_{r\in(\epsilon,1/\epsilon)} \frac{f(x-r\hat{y})}{r^n} r^{n-1} \\
&= \int_{\hat{y}\in S^{n-1}} \Omega(\hat{y}) \int_{r\in(\epsilon,1/\epsilon)} \frac{f(x-r\hat{y})}{r}
\end{aligned}
$$

and

$$
H_k f := \lim_{\epsilon\to 0} H_k^\epsilon
$$

We define the transformations

$$
\begin{aligned}
U_{\hat{y}}^t &: \mathbb{R}^n \to \mathbb{R}^n \\
U_{\hat{y}}^t(x) &:= x + t\hat{y}
\end{aligned}
$$

for $\hat{y} \in S^{n-1}$ and $t \in \mathbb{R}$.

For

$$
f : \mathbb{R}^n \to \mathbb{C}
$$

we have

$$
\begin{aligned}
g_{\hat{y}}^x &: \mathbb{R} \to \mathbb{C} \\
g_{\hat{y}}^x(t) &:= f(U_{\hat{y}}^t(x)) = f(x + t\hat{y})
\end{aligned}
$$

Proposition:

$$
2(H_k^\epsilon f)(x) = \int_{\hat{y}\in S^{n-1}} (H_\epsilon g_{\hat{y}}^x)(0)\Omega(\hat{y})
$$

Before describing this general situation, however, let us establish (1.6). We shall use polar coordinates in order to express certain integrals over $\mathbf{R}^n$: if $h \in L^1(\mathbf{R}^n)$ and $x = rx' \in \mathbf{R}^n$, where $x' \in \Sigma_{n-1}$ and $r = |x| \geq 0$, then

$$\int_{R^n} h(x)\,dx = \int_{\Sigma_{n-1}} \left\{ \int_0^\infty h(rx') r^{n-1}\,dr \right\} dx' \tag{1.8}$$

(dx' is the element of Lebesgue measure on the surface Σ_{n-1}). Thus, since $\Omega(ry') = \Omega(y')$ (Ω being homogeneous of degree 0)

$$2(H_k^\epsilon f)(x) = 2\int_{\epsilon \leq |y| \leq 1/\epsilon} k(y) f(x-y)\,dy$$

$$= 2\int_{\Sigma_{n-1}} \Omega(y') \left\{ \int_\epsilon^{1/\epsilon} \frac{f(x-ry')}{r^n} r^{n-1} dr \right\} dy'.$$

Since $\int_{\Sigma_{n-1}} \Phi(y')\,dy' = \int_{\Sigma_{n-1}} \Phi(-y')\,dy'$, whenever $\Phi \in L^1(\Sigma_{n-1})$, and since $\Omega(-y') = -\Omega(y')$ (by assumption), it follows that the last expression equals

$$\int_{\Sigma_{n-1}} \Omega(y') \left\{ \int_{\epsilon \leq |r| \leq 1/\epsilon} \frac{f(x-ry')}{r} dr \right\} dy' = \int_{\Sigma_{n-1}} \Omega(y')(H_\epsilon g_{y'}^x)(0)\,dy'$$

and (1.6) is established.

-

Proposition:

$$\|H_k^\epsilon f\|_p \leq B_p \|f\|_p$$

From

Proposition:

$$2(H_k^\epsilon f)(x) = \int_{\hat{y} \in S^{n-1}} (H_\epsilon g_{\hat{y}}^x)(0)\Omega(\hat{y})$$

$$\begin{aligned} 2\|H_k^\epsilon f\|_p &= \left(\int_{x \in \mathbb{R}^n} \left| \int_{\hat{y} \in S^{n-1}} (H_\epsilon g_{\hat{y}}^x)(0)\Omega(\hat{y}) \right|^p \right)^{1/p} \\ &\leq \int_{\hat{y} \in S^{n-1}} \left(\int_{x \in \mathbb{R}^n} \left| (H_\epsilon g_{\hat{y}}^x)(0)\Omega(\hat{y}) \right|^p \right)^{1/p} \end{aligned}$$

- Now the integral inside is for a fixed $\hat{y} \in S^{n-1}$, so we write

$$x = s\hat{y} + w$$

for $w \in T_{\hat{y}}S^{n-1} \hookrightarrow \mathbb{R}^n$ and split the integral

$$\begin{aligned}
&\int_{x\in\mathbb{R}^n} \left|(H_\epsilon g^x_{\hat{y}})(0)\Omega(\hat{y})\right|^p \\
=&|\Omega(\hat{y})|^p \int_{w\in T_{\hat{y}}S^{n-1}} \int_{s\in\mathbb{R}} \left|(H_\epsilon g^x_{\hat{y}})(0)\right|^p \\
=&|\Omega(\hat{y})|^p \int_{w\in T_{\hat{y}}S^{n-1}} \int_{s\in\mathbb{R}} \left|\int_{\epsilon\le|t|\le\frac{1}{\epsilon}} \frac{f((s-t)\hat{y}+w)}{t}\right|^p \\
\le&|\Omega(\hat{y})|^p \int_{w\in T_{\hat{y}}S^{n-1}} A_p^p \int_{s\in\mathbb{R}} |f(s\hat{y}+w)|^p \\
\le&A_p^p|\Omega(\hat{y})|^p\|f\|^p_{L^p(\mathbb{R}^n)}
\end{aligned}$$

- Here, we took the estimate from one dimension

  Proposition: For the Hilbert transform

  > **Definition. Hilbert transform on $\mathbb{R}$**
  >
  > For $p \in [0,\infty]$, the **Hilbert transform** is the operator
  >
  > $$\begin{aligned} H : L^p(\mathbb{R}) &\to L^p(\mathbb{R}) \\ f \mapsto Hf &:= \lim_{\epsilon\to 0+} \underbrace{\frac{1}{\pi}\int_{\epsilon\le|t|} \frac{f(s-t)}{t}}_{H_\epsilon f} \end{aligned}$$

  if $1 < p < \infty$ we have

  $$f \in L^p(\mathbb{R}) \implies \|H_\epsilon f\|_p \le A_p\|f\|_p$$

  applied it over lines $\{s\hat{y}+w \mid s\in\mathbb{R}\}$ in $\mathbb{R}^n$ and then averaged it over all $w \perp \hat{y}$: so averaging it over *all lines* in in $\mathbb{R}^n$ in the direction $\hat{y}$.

- Thus

$$2\|H^\epsilon_k f\|_p \le \left(\int_{\hat{y}\in S^{n-1}} |\Omega(\hat{y})|\right) A_p\|f\|_{L^p(\mathbb{R}^n)}$$

  - This further averaged it over all $\hat{y} \in S^{n-1}$, essentially averaging the result over *all lines* in $\mathbb{R}^n$.

- Hence

$$B_p = \frac{1}{2}A_p\|\Omega\|_{L^1(S^{n-1})}$$

  concludes the result.

on the circle

$$\mathbb{R} \curvearrowright S^1$$

by isometries

general transference results

> **Definition. Continuous representation of an amenable group on L^p space of a σ-finite space**
>
> Let G is a locally compact group with the *amenable* property: for any compact $C \subset G$ and $\epsilon > 0$ there exists an open nbd V of identity having finite left Harr measure $\nu(V)$ such that

$$\frac{\nu(VC^{-1})}{\nu(V)} \leq 1+\epsilon$$

Let

$$R: G \to \mathcal{B}(L^p(X,\mu))$$

be a continuous representation of G for (X,μ) a σ-finite measure space.
Let $k \in L^1(G)$ have compact support C and $N_p(k)$ the operator norm of the convolution operator $\|k * -\|_{L^p(G)\to L^p(G)}$ for $p \geq 1$.

We have the **R-convolution with k** operator

$$\begin{aligned} H_k : L^p(X,\mu) &\to L^p(X,\mu) \\ F \mapsto &(H_k F)(x) := \int_{u\in G} k(u)(R_{u^{-1}}F)(x) \end{aligned}$$

For

Definition. Continuous representation of an amenable group on L^p space of a σ-finite space

Let G is a locally compact group with the *amenable* property: for any compact $C \subset G$ and $\epsilon > 0$ there exists an open nbd V of identity having finite left Harr measure $\nu(V)$ such that

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We have the R-convolution with k operator

$$\begin{aligned} H_k : L^p(X,\mu) &\to L^p(X,\mu) \\ F \mapsto &(H_k F)(x) := \int_{u\in G} k(u)(R_{u^{-1}}F)(x) \end{aligned}$$

let there exists an uniform operator bound: $\exists c > 0$ such that

$$\|R_u F\| \leq c\|F\|_p$$

for all $F \in L^p(X,\mu), u \in G$. Then the operator H_k is well-defined and is bounded with

$$\|H_k\|_p \leq c^2 N_p(k)$$

Amos Nevo - Pointwise Ergodic Theorems for Actions of Groups (2006)

Handbook of Dynamical Systems 1, Part B - Hasselblatt and A. Katok (Eds.) - (2006).pdf > page=884

averaging operators of measurable preserving group actions

Definition. LC2 σ-Borel G-space X

Let

- G be a locally compact 2-countable group *(LC2)*
- X *LC2* space on which

$$G \curvearrowright X$$

  acts by a group of homomorphisms
- $m_{X/G}$ be a G-invariant σ-finite Borel measure on X

This gives rise

- to a representation

$$\begin{aligned}\pi : G \curvearrowright L^p(X, m_{X/G})\\ f \overset{\pi_g}{\longmapsto} f \circ g^{-1}\end{aligned}$$

  by isometries of $L^p(X)$
- for $p \in [1, \infty)$

$$G \to \text{Iso}(L^p(X))$$

  is strongly continuous
- For any Borel probability measure μ on G we have **averaging operator**

$$\begin{aligned}P_g := \int_G \pi_g(-) : L^p(X, m_{X/G}) \to L^p(X, m_{X/G})\\ f \overset{\mu}{\mapsto} \int_G f(g^{-1}x)\delta\mu_g\\ = \int_G (\pi_g f)(x)\delta\mu_g\end{aligned}$$

  for $p \in [1, \infty]$, $g \in M$.

  - > $P_g(g)$ is G-invariant?

This is reminiscent of such operators in finite group representation theory!

regular representation

Let $X = G$ and the measure $m = \mu_G$ is the left-invariant Harr measure.

2.1.1. *The regular representation and the action by convolutions.* Consider the case where $X = G$, and the measure $m = m_G$ is left-invariant Haar measure. Then the operators $\lambda(g)f(x) = f(g^{-1}x)$ are isometric in every $L^p(G, m_G)$, and $g \mapsto \lambda(g)$ is the left regular representation. If μ is absolutely continuous with density $d\mu = b(g)\,dm_G(g)$, then the operator $\lambda(\mu)$ is the operator of left convolution by μ:

$$\lambda(\mu)f(x) = \int_G f\left(g^{-1}x\right)b(g)\,dm_G(g) = \mu * f(x).$$

We can similarly consider the right regular representation of G, given by $\rho(g)f(x) = f(xg)$. Of course, in general $\rho(g)$ is *not* an isometric operator in $L^p(G, m_G)$, unless the left Haar measure m_G is also right-invariant, namely unless G is unimodular. Note also that the operator $\rho(\mu)$ is given by $\rho(\mu)f(x) = \int_G f(xg)b(g)\,dm_G(g)$, and is equal to the convolution operator $f * \mu^\vee$ if and only if G is unimodular (here $\mu^\vee(A) = \mu(A^{-1})$). In particular if G is unimodular and the measure μ is symmetric (namely satisfies $\mu^\vee = \mu$), then $\rho(\mu)f = f * \mu$.

ergodic theorems

Definition. Family of averaging operators on a probability G-space

Let G be a LC2 group, $(X, \mathfrak{M}, m)$ be a *probability space*

$$G \to \mathrm{AutMeas}(X, \mathfrak{M}, m)$$

be a *group homomorphism*.

Let $\mu_t, t \in \mathbb{R}_+$ be a family of *probability measures* on G such that

$$\begin{aligned} \mu : \mathbb{R}_+ &\to M(G) \cong \mathcal{C}_0(G)^* \\ t &\mapsto \mu_t \end{aligned}$$

is w^*-continuous. These are regarded as a **family of averaging operators**

$$(\pi(\mu_t)f)(x) := \int_{g\in(G,\mu_t)} f(g^{-1}x)$$

producing a *sampling method* along the orbits Gx.

Usually each μ_t will have some compact support $A_t \subset G$.

Question

We want to establish a *pointwise ergodic theorem* in L^p for families of averaging operators μ_t on G for an ergodic action on X, that is

$$\lim_{t\to\infty} \pi(\mu_t)f(x) = \int_{(X,m)} f$$

for m-almost every $x \in X$ for $f \in L^p(X, m), 1 \le p < \infty$.

> **Bug**
>
> The maximal function associated with a family of averaging operators is "defined" by
>
> $$f \mapsto (M_\mu^* f)(x) := \sup_{t \in \mathbb{R}_+} |\pi(\mu_t) f(x)|$$

growth type of groups

[1]

Let G be a compactly generated, LC2 group, that is

$$V \subset G$$

is a compact set such that

$$\bigcup_{n \in \mathbb{N}} V^n = G$$

where $V^0 := \{i_G\}$.

- Clearly

  $$A, B \subset AB \subset G$$

  thus

  $$k \in \mathbb{N} \implies V^n \subset V^{n+k} = V^n V^k$$

We define the distance

$$|g|_V := \min\{n \mid g \in V^n\}$$

The distance function is inversion invariant $\iff$ V is symmetric $V = V^{-1}$ $\iff$ the function $d_V(g,h) := |g^{-1}h|_V$ is symmetric in the pair (h, g). So we assume $V = V^{-1}$. Then

$$d_V : G \times G \to \mathbb{R}$$

is a left G-invariant metric on G called the **word metric** defined by V.

- G is LC2 $\implies$ there is some $n \in \mathbb{N}$ such that V^n contains an open set
  - by

    > - Clearly
    >
    > $$A, B \subset AB \subset G$$
    >
    >   thus
    >
    > $$k \in \mathbb{N} \implies V^n \subset V^{n+k} = V^n V^k$$

    we have

    $$m_G(V^{n+k}) > 0$$

    and

    $$m_G(V^{n+m}) = m_G(V^n V^m) \leq ? m_G(V^n) m_G(V^m)$$

Pansu's 1983 theorem

In these examples the homogeneity of the metric on G, namely the fact that $B_t = \alpha_t(B_1)$ (where α_t is the dilation automorphism group) immediately implies that $m_G(B_t) = m_G(B_1)t^q$, where q is the homogeneous dimension. Thus in particular the polynomial volume growth for homogeneous metrics is exact. Riemannian metrics are not homogeneous in general, and here, under one further assumption, the following result was established by P. Pansu.

THEOREM 4.4 [123]. *Let G be a simply connected nilpotent Lie group admitting a lattice subgroup Γ. Then*

(1) *G has exact polynomial volume growth w.r.t. any Riemannian metric on G which is Γ-invariant.*
(2) *Any lattice Γ in G has exact polynomial volume growth with respect to any word metric.*

Pansu's theorem has the following two consequences [123] (see also the discussion in [42, Chapter VII.C]).

COROLLARY 4.5.

(1) *Every discrete group containing a finitely generated nilpotent group of finite index has exact polynomial volume growth with respect to any word metric. Indeed, by a well-known result of Malcev, Γ can be embedded as a lattice in a simply connected nilpotent Lie group.*
(2) *Every discrete group Γ with polynomial volume growth has exact polynomial volume growth with respect to any word metric. Indeed, by Gromov's theorem* [63], *Γ is virtually nilpotent, namely contains a nilpotent subgroup of finite index.*

We note that a sharper form of Pansu's theorem has been established by M. Stoll [137], namely that $|\#B_t - Ct^q| \leqslant Bt^{q-1}$ for 2-step nilpotent groups.

- [123] https://www.cambridge.org/core/services/aop-cambridge-core/content/view/9352E615E18472BAE919B24621420F9B/S0143385700002054a.pdf/croissance-des-boules-et-des-geodesiques-fermees-dans-les-nilvarietes.pdf
    - Croissance des boules et des géodésiques fermées dans les nilvariétés | Ergodic Theory and Dynamical Systems | Cambridge Core
- [15] missing??
- generalization, 2007? - https://arxiv.org/pdf/0704.0095#page=2.0
- asymptotics, 2012 - https://arxiv.org/pdf/1204.1613#page=2.05
- [137] https://sci-hub.se/10.1112/s0024610798006371
- https://ep455.user.srcf.net/pdfs/nilpoly.pdf
- [42] (Chicago Lectures in Mathematics) Pierre de la Harpe - Topics in Geometric Group Theory-The University of Chicago Press (2000), p.204
- (Grundlehren der mathematischen Wissenschaften) Martin R. Bridson, André Häfliger - Metric Spaces of Non-Positive Curvature -Springer (2009), p.95
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- https://www.imo.universite-paris-saclay.fr/~pierre.pansu/Pansu_Pisa_jun23_minicourse_beamer.pdf
- https://cvgmt.sns.it/media/doc/paper/5339/sub-Riem_notes.pdf#page=229.65
  - Topics in geometry: Nilpotent groups and subriemannian geometry
  - SubRiemannian geometry course - YouTube

Theorem 11.4.14 (Quantitative Pansu Theorem; [BL13]). *Let G be a simply connected s-step nilpotent Lie group equipped with a left-invariant sub-Finsler metric ρ_1. With respect to some compatible linear grading, consider the Pansu limit metric ρ_0 on G and the contracted metrics ρ_ϵ as in* (11.4.6)*. For every compact set $K \subseteq G$ there is a constant $C > 0$ such that*

$$|\rho_\epsilon(p,q) - \rho_0(p,q)| \leq C\epsilon^{1/s}, \qquad \forall p,q \in K, \forall \epsilon \in (0,1). \tag{11.4.15}$$

11.4.1 Pansu asymptotic structure

Given the nilpotent Lie algebra $\mathfrak{g}$, we consider the associated Carnot algebra $\mathfrak{g}_\infty$ of G, as in Definition 8.2.25. This is a stratifiable algebra, and $\mathfrak{g}/[\mathfrak{g},\mathfrak{g}]$ is the first stratum of a stratification. Consider G_∞ to be the nilpotent simply connected Lie group with Lie algebra $\mathfrak{g}_\infty$, by recalling the existence by Birkhoff Theorem; see Proposition 8.4.7. We shall put a Carnot metric on it. Consider the projection $\pi : \mathfrak{g} \to \mathfrak{g}/[\mathfrak{g},\mathfrak{g}]$ modulo $[\mathfrak{g},\mathfrak{g}]$. Since Δ_{1_G} generates the whole Lie algebra and since $\mathfrak{g}/[\mathfrak{g},\mathfrak{g}]$ is an abelian Lie algebra, we have $\pi(\Delta_{1_G}) = \mathfrak{g}/[\mathfrak{g},\mathfrak{g}]$. We define a norm $\|\cdot\|_\infty$, on $\mathfrak{g}/[\mathfrak{g},\mathfrak{g}]$, which we name *Pansu limit norm*, by imposing $B_{\|\cdot\|_\infty}(0,1) = \pi(B_{\|\cdot\|}(0,1))$. Let d_∞ be the subFinsler distance associated with $(G_\infty, \mathfrak{g}/[\mathfrak{g},\mathfrak{g}], \|\cdot\|_\infty)$. Thus (G_∞, d_∞) is a Carnot group. We call d_∞ the *Pansu limit metric* or the *asymptotic metric*.

Pansu Theorem 11.1.8 will follow from a more quantitative version as obtained in [BL13]:

Theorem 11.4.1 ([Pan83, BL13]; see Theorem 11.4.14). *Let (G,d) be a nilpotent simply connected Lie group equipped with a sub-Finsler left-invariant metric. Then, the associated Carnot group G_∞ equipped with the Pansu limit metric d_∞ is the asymptotic cone of (G,d) and there is a set-wise identification between G and G_∞ such that*

$$\Big|d(p,q) - d_\infty(p,q)\Big| = O\left(\max\{d(1,p), d(1,q)\}^{1-1/s}\right), \qquad \textit{as } p,q \to \infty, \tag{11.4.2}$$

where s denotes the nilpotency step of G.

Theorem 11.1.8 (Pansu; [Pan83]; see Theorem 11.4.1)**.** *Let G be a nilpotent simply connected Lie group equipped with a left-invariant sub-Finsler distance. Then, the asymptotic cone of G exists and is a sub-Finsler Carnot group.*

appendix

[2]

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1. https://www.math.uchicago.edu/~may/VIGRE/VIGRE2009/REUPapers/Lim.pdf ↩
2. https://arxiv.org/pdf/2105.02090 ↩

Uri Bader - A course in linear groups and ergodic theory

$SL(2,\mathbb{R}) \curvearrowright \mathbb{R}^2$

Consider

$$\underbrace{SL(2,\mathbb{R})}_{=:G} \curvearrowright \underbrace{\mathbb{R}^2}_{=:X}$$

There are two orbits

$$\{0\}$$

and

$$\mathbb{R}^2 \setminus \{0\}$$

(disjoint of course).

By orbit-stabilizer theorem

$$\mathbb{R}^2 \setminus \{0\} \cong \frac{G}{\mathcal{U}}$$

where stabilizer is the subgroup

$$\mathcal{U} := \mathrm{stab}_G \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \left\{ \begin{bmatrix} 1 & * \\ 0 & 1 \end{bmatrix} \right\} = \exp \left\{ \begin{bmatrix} 0 & * \\ 0 & 0 \end{bmatrix} \right\} \cong (\mathbb{R}, +)$$

$SL(2,\mathbb{R})$-maps

> **Question**
>
> Find all G-invariant maps
>
> $$\phi : \mathbb{R}^2 \to \mathbb{R}^2$$
>
> that is
>
> $$\forall g \in G, x \in \mathbb{R}^2, \phi(gx) = g\phi(x)$$

Because G-orbits will go to G-orbits under such a map we conclude

$$\phi(0) = 0$$

But if we choose where one point of a orbit will go, then we know where the other elements will go

$$\underbrace{g\phi(x)}_{\text{known}} = \underbrace{\phi(gx)}_{\text{unkown}}$$

But to know

$$\phi \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

we observe

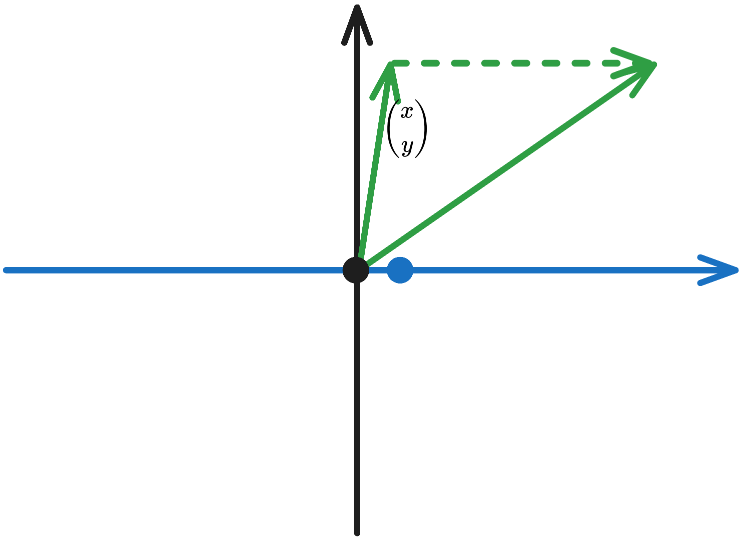
$$g \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \implies \phi \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \phi(g \begin{pmatrix} 1 \\ 0 \end{pmatrix}) = g\phi \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

means

$$\mathcal{U} \subset \mathrm{stab}_G(\phi \begin{pmatrix} 1 \\ 0 \end{pmatrix})$$

So we must look at *the orbit structure of $\mathcal{U} \curvearrowright \mathbb{R}^2$* to know what point does it *stabilizes*:

$$\begin{bmatrix} 1 & \alpha \\ 0 & 1 \end{bmatrix} \in \mathcal{U} \implies \begin{bmatrix} 1 & \alpha \\ 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \alpha \begin{pmatrix} y \\ 0 \end{pmatrix}$$



flow

The dynamics is "tame".

So

$$\mathcal{U} \subset \mathrm{stab}_G(\phi \begin{pmatrix} 1 \\ 0 \end{pmatrix}) \iff \phi \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \alpha \\ 0 \end{pmatrix}$$

Then

$$\phi(y) = \phi(g \begin{pmatrix} 1 \\ 0 \end{pmatrix}) = g\phi \begin{pmatrix} 1 \\ 0 \end{pmatrix} = g \begin{pmatrix} \alpha \\ 0 \end{pmatrix} = \alpha g \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \alpha y$$

Thus, every G-invariant map $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ is a scaling

$$\phi(x) = \alpha x$$

by some $\alpha \in \mathbb{R}$, which is invertible if $\alpha \in \mathbb{R}^\times$.

Exercise

$$\mathrm{Aut}G\mathsf{Set}\left(\frac{G}{H}\right) \cong \frac{N_G(H)}{H}$$

For our case $H = \mathcal{U}$

$$N_G(\mathcal{U}) = \left\{ \begin{bmatrix} * & * \\ 0 & * \end{bmatrix} \right\} \cong \mathbb{R}^\times \times \mathbb{R}$$

and

$$\frac{N_G(\mathcal{U})}{\mathcal{U}} \cong \mathbb{R}^\times$$

$SL(2,\mathbb{Z})$-maps

We may ask

Question

Find all Borel maps

$$\mathbb{R}^2 \to \mathbb{R}^2$$

that commutes with $\Gamma := SL(2,\mathbb{Z})$ defined upto Lebesgue measure-zero sets.

whose answer is

Every Borel measurable $SL(2,\mathbb{Z})$-map $\mathbb{R}^2 \to \mathbb{R}^2$ is essentially a scaling upto Lebesgue measure-zero sets.

But what is special about Γ? The fact that it is a "*lattice*" in G that is:

> **Proposition:** $\Gamma < G$ is discrete and the space
>
> $$G \curvearrowright \frac{G}{\Gamma}$$
>
> has a G-invariant probability measure.

The subgroup

$$A := \left\{ \begin{bmatrix} \lambda & \\ & \lambda^{-1} \end{bmatrix} \mid \lambda \in \mathbb{R}^\times \right\} < G = SL(2,\mathbb{R})$$

which acts by $\begin{bmatrix} -1 & \\ & -1 \end{bmatrix}$ along with the flow

$$\left\{ \exp\left(\alpha \begin{bmatrix} 1 & \\ & -1 \end{bmatrix} \right) \middle| \alpha \in \mathbb{R} \right\}$$

flow of A

This dynamics is "*tame*" as in the orbits are "locally closed".

"Tame dynamics"	"Wild dynamics"
action by $\mathcal{U}, A$ above on $\mathbb{R}^2$	irrational torus rotations
	$\Gamma \curvearrowright \frac{G}{\mathcal{U}}$
	$\mathcal{U} \curvearrowright \frac{G}{\Gamma}$

There are no $SL(2,\mathbb{R})$-invariant metric on $\mathbb{R}^2$

Every continuous

$$\begin{aligned} d : \mathbb{R}^2 \times \mathbb{R}^2 \to [0, \infty) \\ d(x, y) = d(y, x) \\ d(x, x) = 0 \end{aligned}$$

satisfying triangle inequality which is $SL(2, \mathbb{R})$-invariant, that is,

$$g \in SL(2, \mathbb{R}) \implies d(gx, gy) = d(x, y)$$

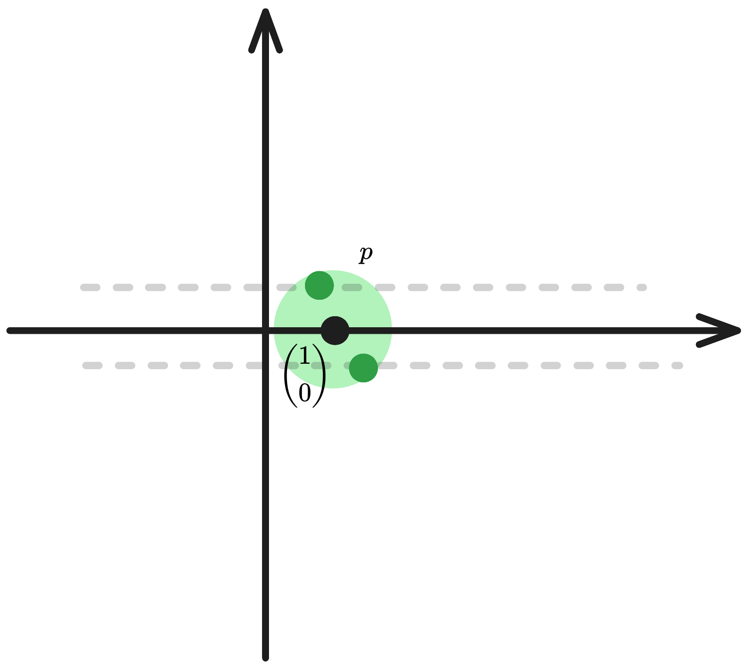
is trivial, as in $d \equiv 0$.

Fix $\epsilon > 0$.

- The restriction $d(\begin{pmatrix}1\\0\end{pmatrix}, -) : \mathbb{R}^2 \to [0, \infty)$ is continuous so preimage of $(0, \epsilon)$ contains a standard ball say B_ϵ.
- Because $\mathcal{U}$ fixes $\begin{pmatrix}1\\0\end{pmatrix}$

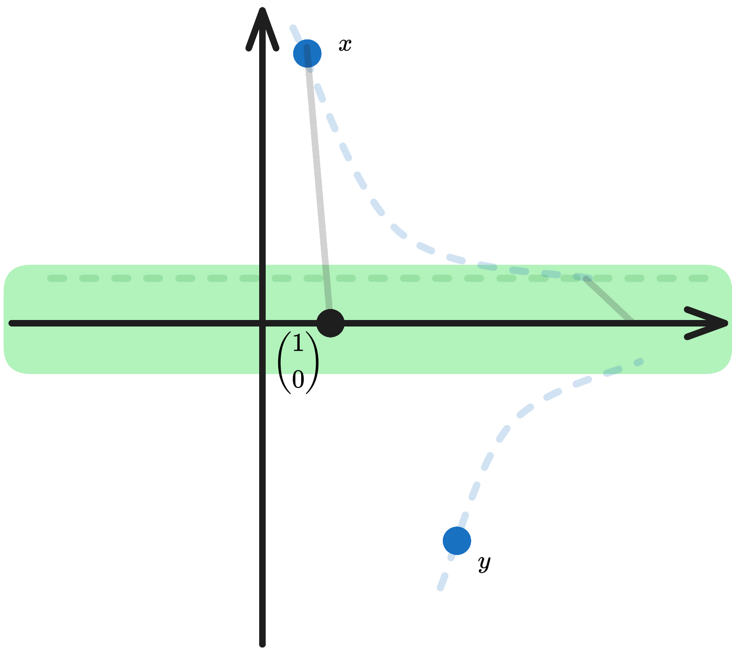
  $$d(gp, \begin{pmatrix}1\\0\end{pmatrix}) = d(p, \begin{pmatrix}1\\0\end{pmatrix}) < \epsilon$$

  for any $p \in B_\epsilon$.

-
- Thus the $d(\begin{pmatrix}1\\0\end{pmatrix}, -)$-preimage of $(0, \epsilon)$ actually contains a horizontal strip $U_\epsilon := \mathbb{R} \times (-\epsilon, \epsilon)$.
- Now pick $x, y \in \mathbb{R}^2$ and flow them along A so that they both reach the U_ϵ, so

  $$gx \in U_\epsilon$$



-
- By triangle-inequality

$$d(x, \begin{pmatrix}1\\0\end{pmatrix}) = d(gx, g\begin{pmatrix}1\\0\end{pmatrix}) \leq d(gx, \begin{pmatrix}1\\0\end{pmatrix}) + d(g\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}1\\0\end{pmatrix}) \leq 2\epsilon$$

- By triangle-inequality

$$d(x,y) \leq d(x, \begin{pmatrix}1\\0\end{pmatrix}) + d(y, \begin{pmatrix}1\\0\end{pmatrix}) = 4\epsilon$$

- So

$$d(x,y) < 4\epsilon$$

  for all $\epsilon > 0$. Thus $d \equiv 0$.

Action of $\mathcal{U}$ on G/Γ is ergodic

Corollary of

Every continuous

$$\begin{aligned} d : \mathbb{R}^2 \times \mathbb{R}^2 \to [0,\infty) \\ d(x,y) = d(y,x) \\ d(x,x) = 0 \end{aligned}$$

satisfying triangle inequality which is $SL(2,\mathbb{R})$-invariant, that is,

$$g \in SL(2,\mathbb{R}) \implies d(gx,gy) = d(x,y)$$

is trivial, as in $d \equiv 0$.

: The action of $\mathcal{U}$ on G/Γ is ergodic

Definition. **Ergodic actions**

An action $G \curvearrowright X$ with an invariant measure class is **ergodic** if either of these is true

- $\forall A \subseteq X$ measurable and invariant A is either a null set or a full measure set
- $L^\infty(X)^G$ is one dimensional

- (if X has a G-invariant probability measure) $L^2(X)^G$ is one dimensional

> We observe the action of G on G/Γ is ergodic because the action is transitive?

Let X is a G-space, $\mathcal{U} < G$ and $\Gamma < G$ is a lattice. Then we have a G-invariant measure on G/Γ so we may consider the L^2 space on it. Then

$$L^2(G/\Gamma)^{\mathcal{U}} \cong \mathrm{Hom}_G\left(\frac{G}{\mathcal{U}}, L^2(G/\Gamma)\right)$$

- **Every continuous**

$$\begin{aligned} d : \mathbb{R}^2 \times \mathbb{R}^2 &\to [0, \infty) \\ d(x, y) &= d(y, x) \\ d(x, x) &= 0 \end{aligned}$$

satisfying triangle inequality which is $SL(2, \mathbb{R})$-invariant, that is,

$$g \in SL(2, \mathbb{R}) \implies d(gx, gy) = d(x, y)$$

is trivial, as in $d \equiv 0$.

implying

$$\mathrm{Hom}_G\left(\frac{G}{\mathcal{U}}, L^2(G/\Gamma)\right) \cong L^2(G/\Gamma)^G \cong \mathbb{C}$$

by

> We observe the action of G on G/Γ is ergodic because the action is transitive?

Ergodic group actions

Definition. Ergodic actions

An action $G \curvearrowright X$ with an invariant measure class is **ergodic** if either of these is true

- $\forall A \subseteq X$ measurable and invariant A is either a null set or a full measure set
- $L^\infty(X)^G$ is one dimensional
- (if X has a G-invariant probability measure) $L^2(X)^G$ is one dimensional

Let X is a G-space, $\mathcal{U} < G$ and $\Gamma < G$ is a lattice. Then we have a G-invariant measure on G/Γ so we may consider the L^2 space on it. Then

$$L^2(G/\Gamma)^{\mathcal{U}} \cong \mathrm{Hom}_G\left(\frac{G}{\mathcal{U}}, L^2(G/\Gamma)\right)$$

duality principle

$$\mathrm{Hom}_{\Gamma}\left(\frac{G}{\mathcal{U}}, \mathbb{R}^{2}\right) \rightarrow \mathrm{Hom}_{\mathcal{U}}\left(\frac{G}{\Gamma}, \mathbb{R}^{2}\right)$$
$$\Phi \mapsto \tilde{\Phi}(g) := g\Phi\left(g^{-1}\right)?$$

Einsiedler and Ward - Ergodic Theory

#book/series

- Einsiedler and Ward - Ergodic Theory (2011).pdf
- Entropy | Books
- Homogeneous | Books

Sigurdur Helgason - Groups and geometric analysis

§1. Harmonic Analysis on Homogeneous Spaces

1. General Problems

Let X be a locally compact Hausdorff space acted on transitively by a locally compact group G. We assume G leaves invariant a positive measure μ on X. Let T_X denote the (unitary) representation of G on the Hilbert space $L^2(X)$ defined by $(T_X(g)f)(x) = f(g^{-1} \cdot x)$ for $g \in G$,

Chapter 1: Integral geometry and Radon transforms

Invariant measures on coset spaces

Definition. Invariant forms

Let M be a smooth manifold and

$$\Phi : M \to M$$

be a diffeomorphism. A differential form $\omega \in \Omega(M)$ is **Φ-invariant** if

$$\Phi^* \omega = \omega$$

Definition. Left and right invariant forms on a Lie group

Let G be a Lie group with Lie algebra $\mathfrak{g}$. A differential form $\omega \in \Omega(G)$ is **left-invariant** if

$$x \in G \implies L_x^* \omega = \omega$$

and **right-invariant** if

$$x \in G \implies R_x^* \omega = \omega$$

Let $X_1, \ldots, X_n$ be a basis of $\mathfrak{g}$ (left-invariant vector fields). Then we have n many 1-forms

$$\omega_1, \ldots, \omega_n$$

such that

$$\omega_i(X_j) = \delta_{ij}$$

These 1-forms are left-invariant and thus

$$\lambda := \omega_1 \wedge \cdots \wedge \omega_n$$

is also a left-invariant n-form on G and it is the unique left-invariant n-form on G up to a constant factor.

Radon transform on $\mathbb{R}^n$

Let

$$\boldsymbol{P}(\mathbb{R}^{n*}) \cong \mathbb{R}\boldsymbol{P}^{n-1}$$

be the compact smooth $n-1$-manifold of all hyperplanes in $\mathbb{R}^n$ passing through origin.

Each hyperplane ξ can be written

$$\xi = \{x \in \mathbb{R}^n \mid \langle x, \hat{n} \rangle = \alpha\}$$

This produces a *surjective* map onto the space of all hyperplanes $\boldsymbol{Pl}^n$ in $\mathbb{R}^n$

$$\begin{aligned} S^{n-1} \times \mathbb{R} &\to \boldsymbol{Pl}^n \\ (\hat{n}, \alpha) &\mapsto \xi = \{x \in \mathbb{R}^n \mid \langle x, \hat{n} \rangle = \alpha\} \end{aligned}$$

that is a smooth double covering(?) as

$$(\hat{n}, \alpha), (-\hat{n}, -\alpha)$$

produce the same plane. Thus we identify

$$\mathcal{C}^{\bullet}(\boldsymbol{Pl}^n) \leftrightarrow \{f \in \mathcal{C}^{\bullet}(S^{n-1} \times \mathbb{R}) \mid f(\hat{n}, \alpha) = f(-\hat{n}, -\alpha)\}$$

Definition. Radon transform and its dual

Let

$$f : \mathbb{R}^n \to \mathbb{C}$$

be integrable on each hyperplane in $\mathbb{R}^n$. Let $\boldsymbol{P}^n$ denote the compact smooth n-manifold of all hyperplanes in $\mathbb{R}^n$. The **Radon transform** of f is

$$\begin{aligned} J(f) : \boldsymbol{P}^n &\to \mathbb{C} \\ \xi &\mapsto \hat{J}(f)(\xi) := \int_{(\xi, m_{\mathbb{R}^2})} f \end{aligned}$$

and the **dual Radon transform** of $\phi \in \mathcal{C}(\boldsymbol{P}^n)$

$$\begin{aligned} \mathbb{R}^n &\to \mathbb{C} \\ x &\mapsto \check{J}(\phi)(x) := \int_{x \in \xi} \phi \end{aligned}$$

Amie Wilkinson - Geometry, Dynamics, and Rigidity

#book/draft

DGRBookCurrent.pdf, DGRBookCurrent.pdf

references

Karen Butt - Entropies of negatively curved surfaces

Midwest 25 Summer School: Karen Butt - Entropies of negatively curved surfaces

#talk

Abstract

Speaker: Karen Butt
Title: Entropies of negatively curved surfaces
Abstract: I will discuss two notions of entropy for the geodesic flow: the topological entropy and the measure-theoretic entropy with respect to the Liouville measure. How these dynamical invariants relate to the underlying Riemannian metric has long been of interest. For negatively curved surfaces, Katok proved that equality of the Liouville and topological entropies characterizes metrics of constant negative curvature. In this setting, Manning asked how these two entropies vary along the normalized Ricci flow; more specifically, he proved the topological entropy is monotonic. The main result of this talk, joint with Erchenko, Humbert, and Mitsutani, is that the same is true for the Liouville entropy. In addition to geometric and dynamical methods, we use microlocal analysis to differentiate the Liouville entropy with respect to a conformal perturbation of the metric.

Karen Butt #people

[2504.07290] Monotonicity of the Liouville entropy along the Ricci flow on surfaces

Quote

Hi Rupadarshi,

Yes, you can check out this book (still in progress) by Amie Wilkinson.

See, section 2.3 in the introduction for a discussion of entropy rigidity, and there's a whole chapter on it later on. I haven't read the book carefully, but I don't think you need to read the entire book if you are interested in the entropy rigidity specifically. Maybe start in that chapter and work backwards as needed.

Karen

CHAPTER 1

Introduction

The goal of this text is to illustrate how tools from dynamical systems can be brought to bear on rigidity questions in Riemannian geometry, in particular questions about:

- the lengths of closed geodesics in negatively curved surfaces,
- metrics on the torus without conjugate points, and
- numerical invariants of geodesic flows for negatively curved manifolds.

We will present three results that give insight into these topics: The Otal–Croke proof of Marked length spectrum rigidity for surfaces [**247, 83**], the Burago–Ivanov proof of the Hopf conjecture on tori [**64**], and A. Katok's proof of the entropy rigidity conjecture on surfaces [**184**]. We state these results later in this chapter, in Section 2.

First we put these results in a broader context, with an (admittedly personal) view of the role of rigidity in dynamics and geometry. Most of the technical material presented in the next section will be revisited more formally in later chapters, and we have referenced this material accordingly. We have chosen to focus much of this discussion through the lens of surfaces, as many of the basic dynamical and geometric themes emerge already in this restricted setting.

Chapter 11

Appendix to Part 3: the role of microlocal analysis in local geometric rigidity

Questions about geodesics and their behavior in manifolds have long been approached using tools from analysis, particularly microlocal analysis. Dating back to classical studies on Zoll surfaces–compact surfaces whose geodesics are all closed and of equal length–analytical techniques by Israel Gelfand, Alan Weinstein, Guillemin and Kazhdan (e.g. [**313, 153, 149**] laid important foundations. Subsequent influential work of Christopher Croke, Vladimir Sharafutdinov, Gabriel Paternain, Marco Sasso, Gunther Uhlmann and others (e.g. [**89, 250**] has significantly advanced our understanding of geometric inverse problems, building upon classical transformations introduced by Funk and Radon (see also the discussion at the beginning of Section 3).

At the same time, methods from microlocal analysis have found remarkable applications in both classical and quantum dynamics. Nalini Anantharaman's work [**9**] in quantum dynamics, complemented by related research by Carlangelo Liverani and Viviane Baladi [**25**] on anisotropic Sobolev spaces, has demonstrated the power of microlocal analysis in establishing dynamical properties, both quantum and classical, such as quantum unique ergodicity and exponential decay of correlations for hyperbolic systems. Further extensions of these methods by Fréderic Faure and Johannes Sjöstrand [**113**], originating in the study of quantum chaos, have brought these tools to the state of the art.